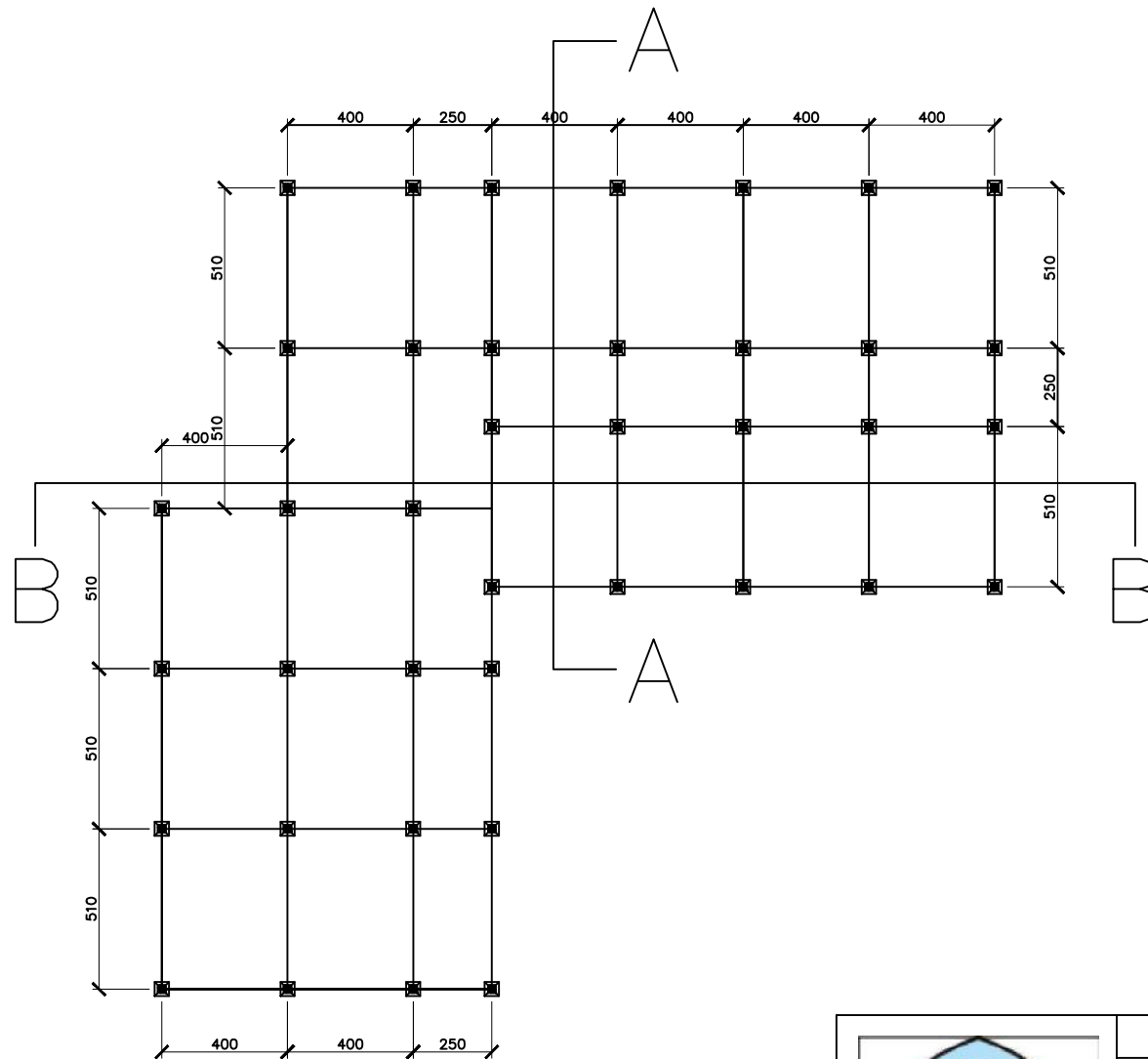




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NAMA KELOMPOK : Kelompok 1

ANGKATAN/KELAS : 2020/3A

DOSEN : Aryati Indah.K.ST.,MT

ASISTEN DOSEN : Ari al'fiansyah

FUNGSI BANGUNAN : -

NAMA GAMBAR : Denah Atap

Tanggal

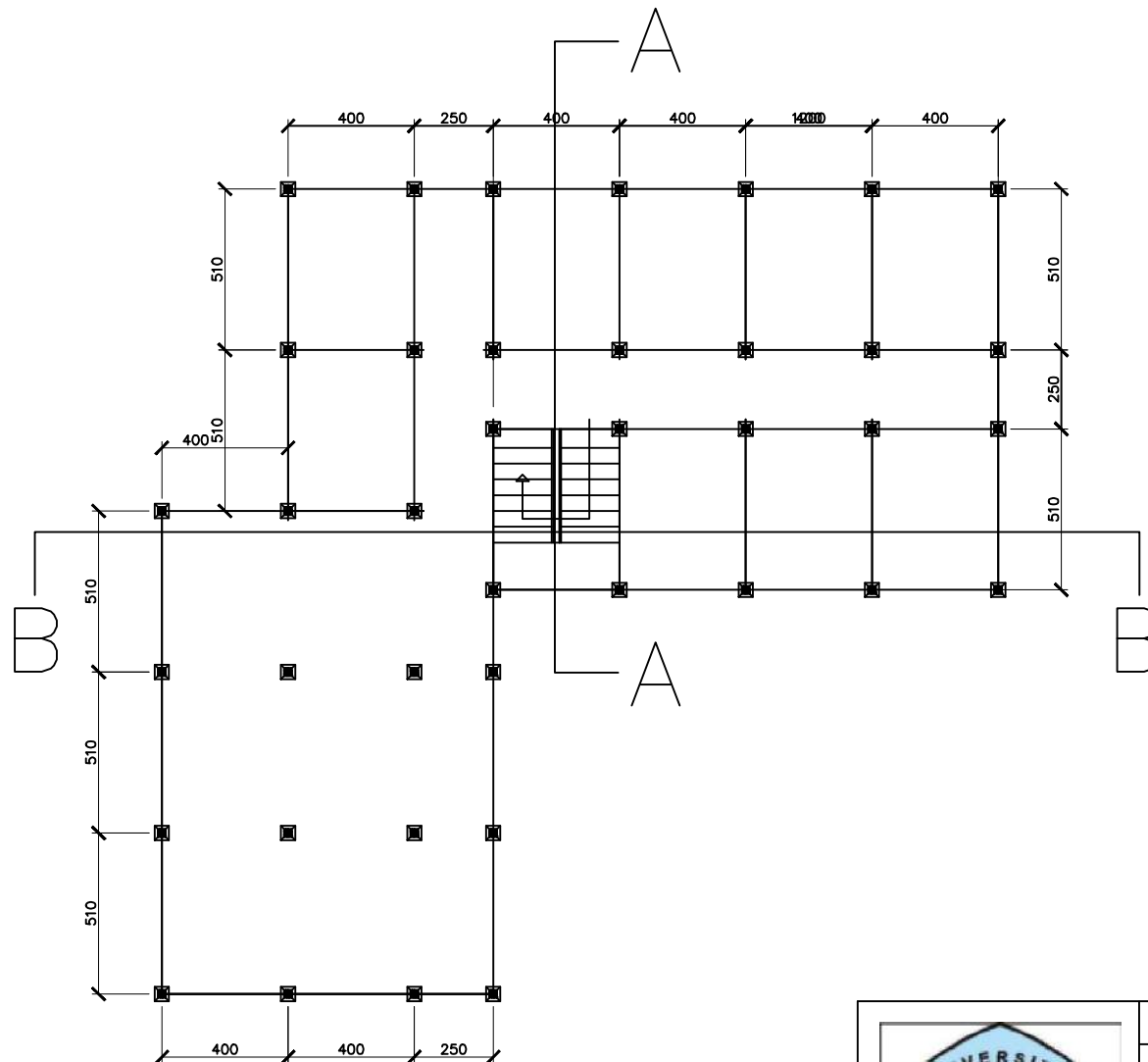
Skala

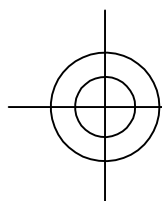
No.Draw

31 Juli 2023

1 : 100

01

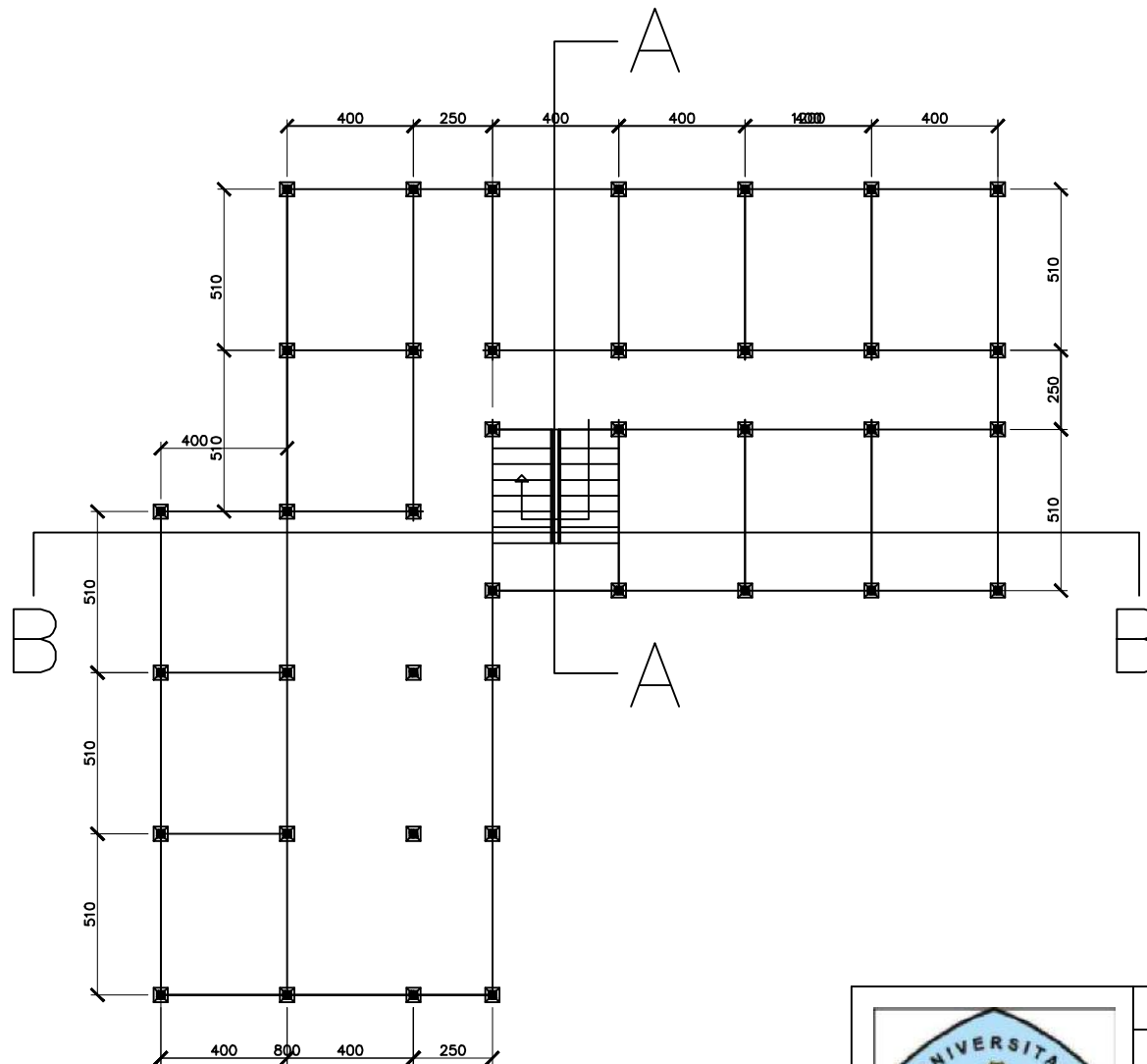


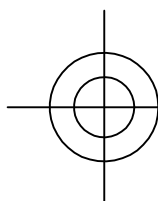

DENAH LT. 1
 SKALA 1 : 100



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 (TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR : Denah Lantai 1		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	02

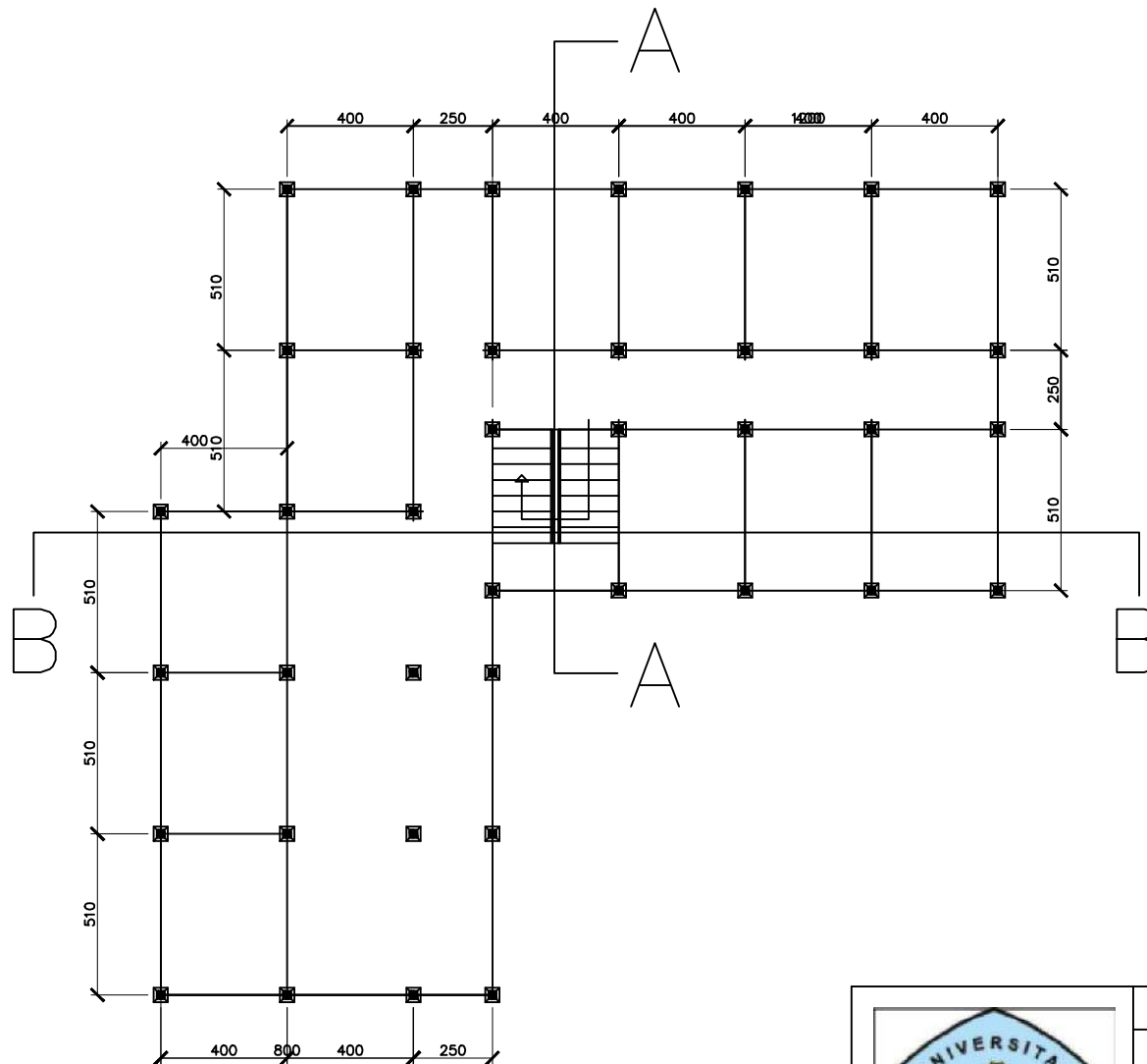


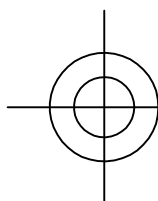

DENAH LT. 2
 SKALA 1 : 100



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 (TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR : Denah Lantai 2		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	03

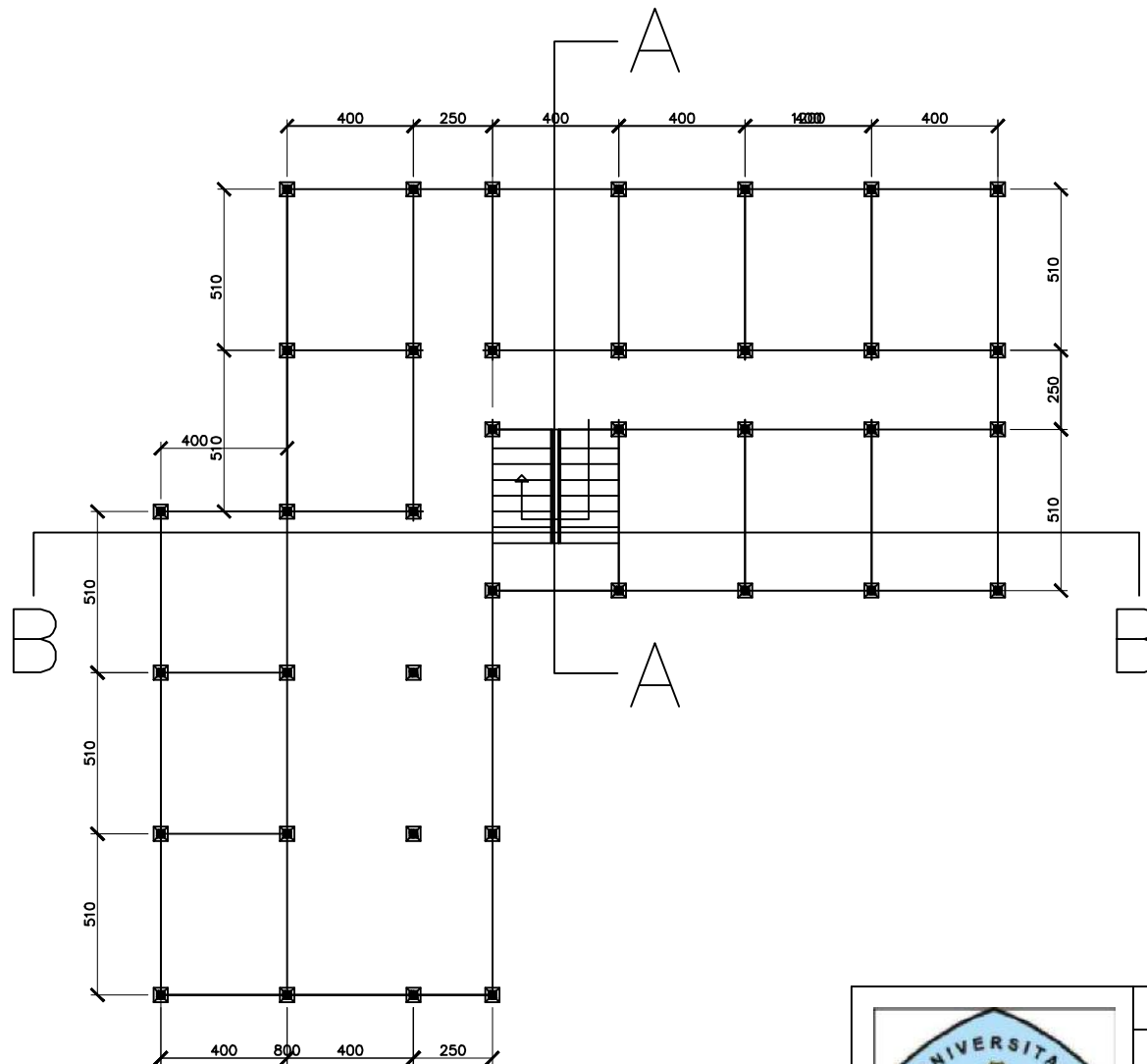


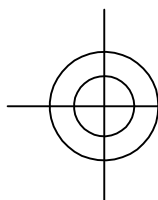

DENAH LT. 3
 SKALA 1 : 100



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 (TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR : Denah Lantai 3		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	04

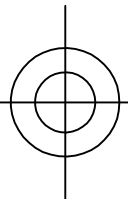



DENAH LT. 4
 SKALA 1 : 100



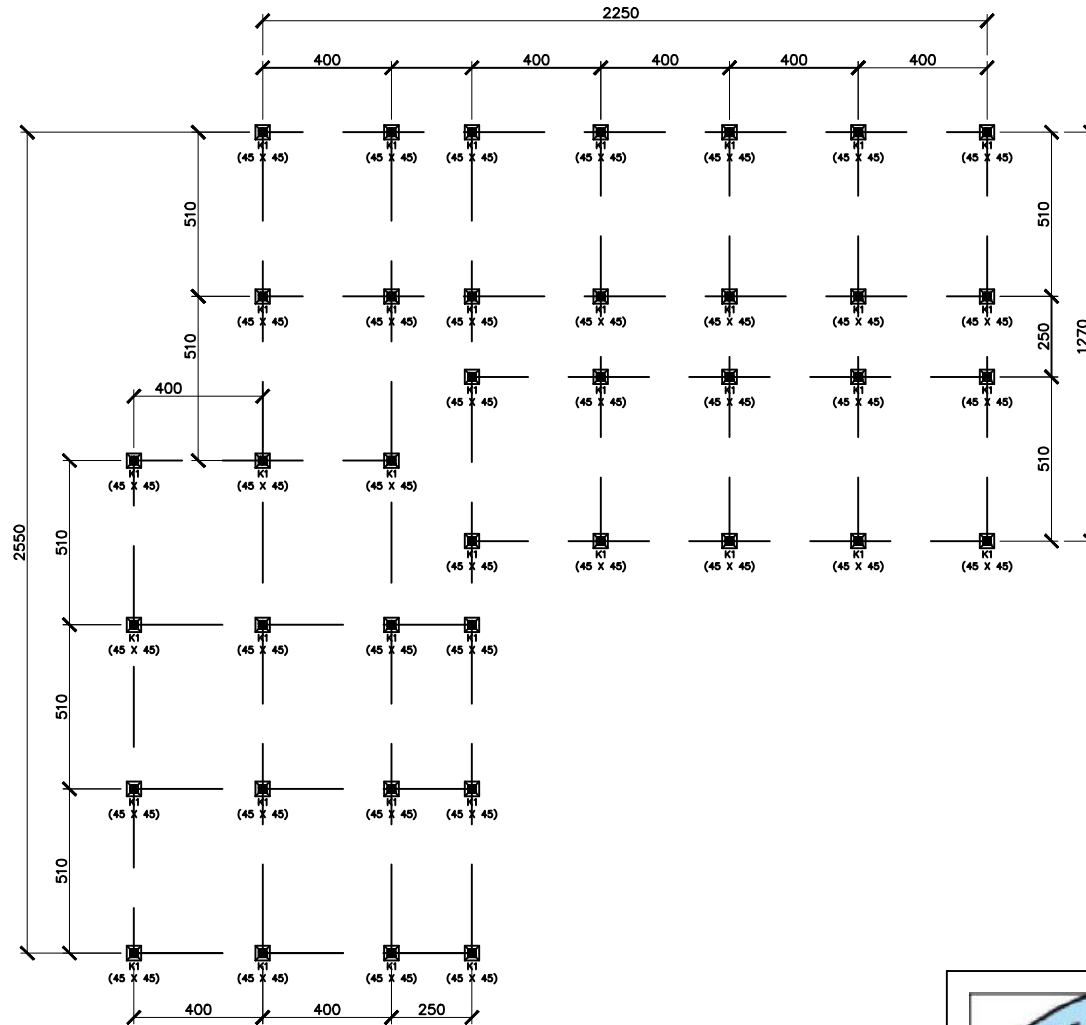
UNIVERSITAS
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 (TEKNIK SIPIL)

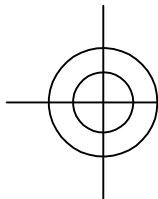
NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR : Denah Lantai 4		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	05



SKALA 1 : 100

Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	06

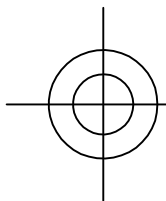
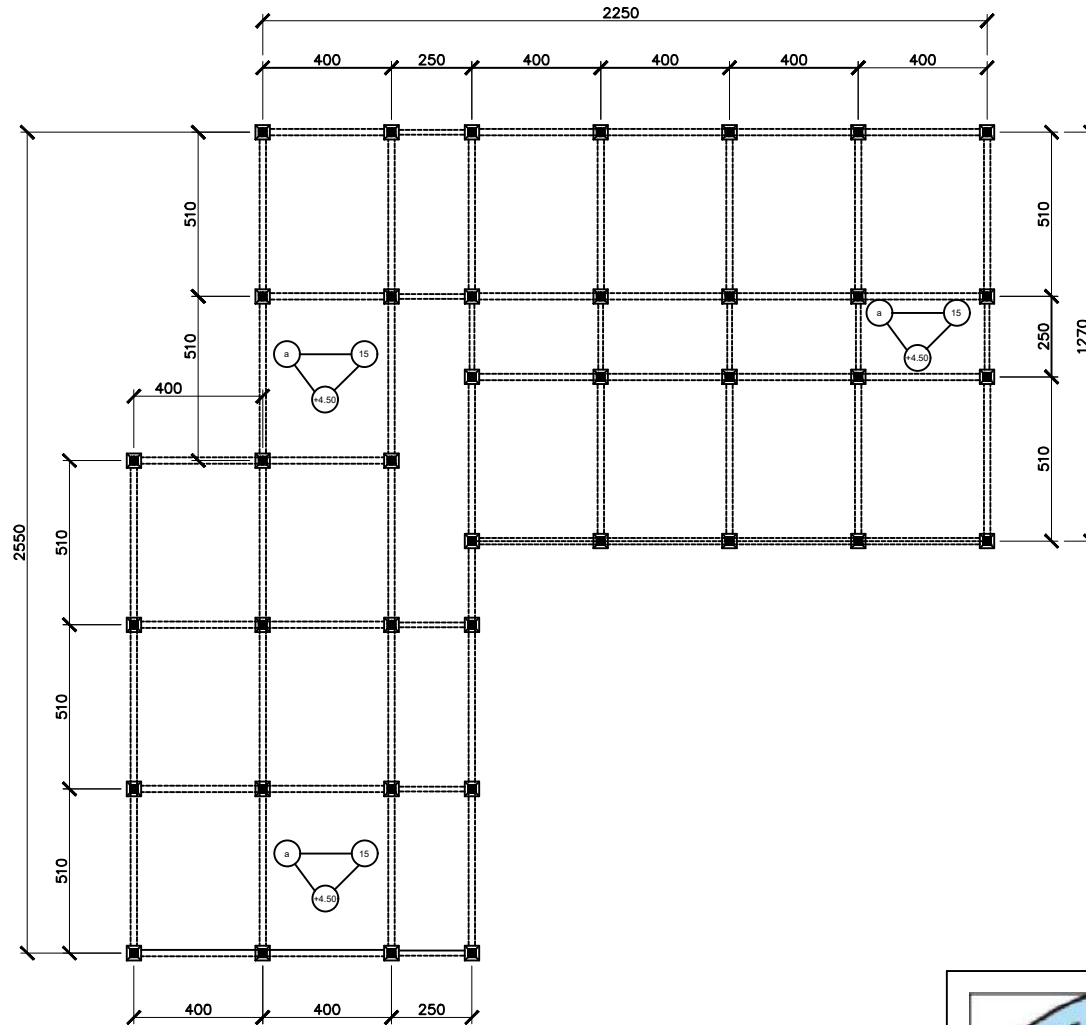



 Floor 1-2 Plan Column
 SKALA 1 : 100



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 (TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : -		
NAMA GAMBAR : Denah Kolom lantai 1-2		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	07



Floor Plan Slab
SKALA 1 : 100



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(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1

ANGKATAN/KELAS : 2020/3A

DOSEN : Aryati Indah.K.ST.,MT

ASISTEN DOSEN : Ari al'fiansyah

FUNGSI BANGUNAN : -

NAMA GAMBAR : Denah Slab

Tanggal

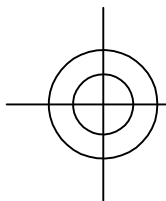
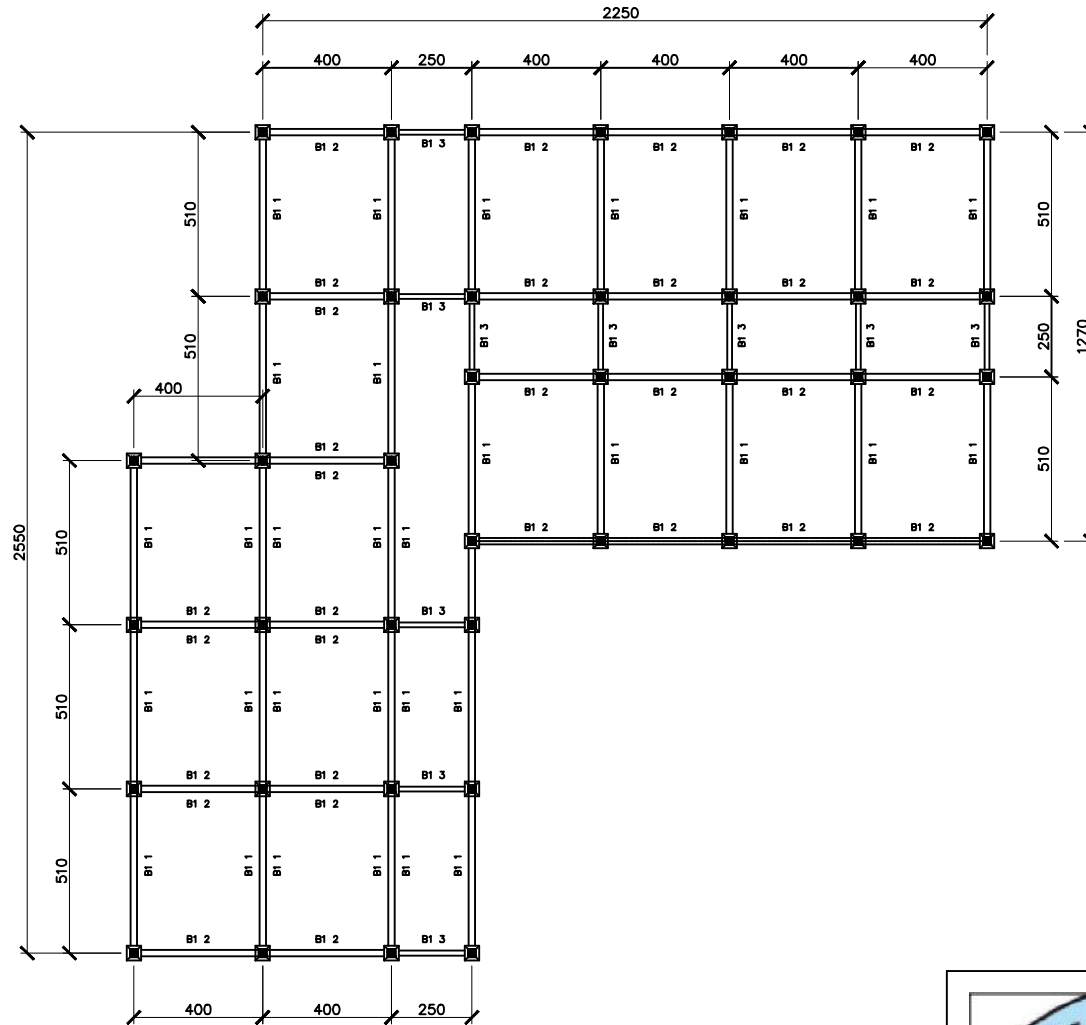
Skala

No.Draw

31 Juli 2023

1 : 100

08

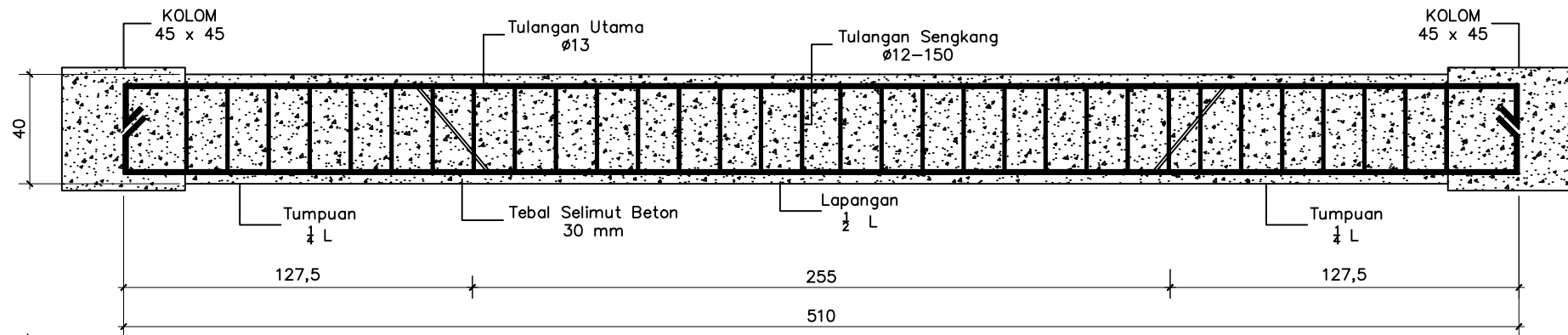


Floor Plan Beam
SKALA 1 : 100

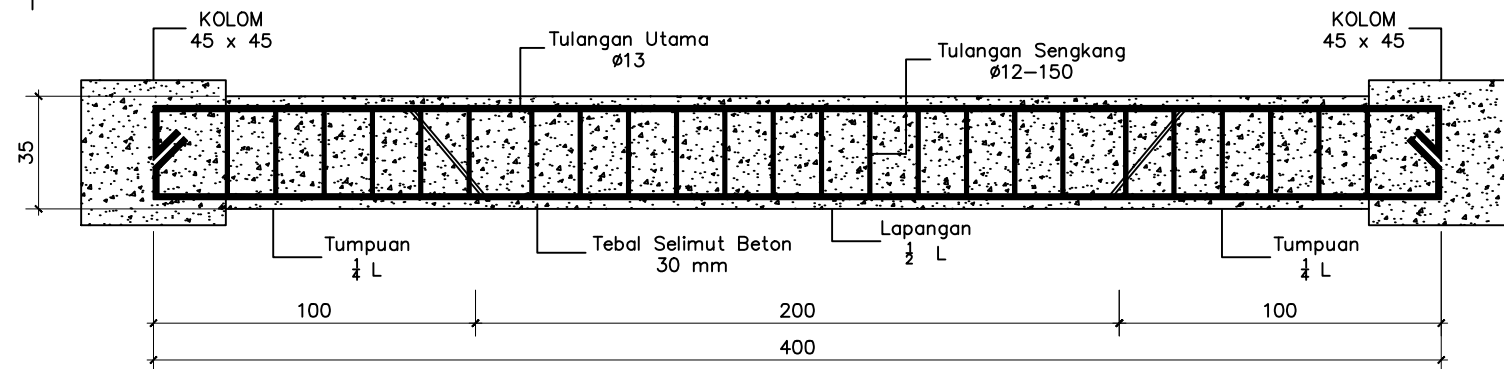


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NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR : Denah Balok		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	09



Potongan balok induk Arah Y
SKALA 1 : 100

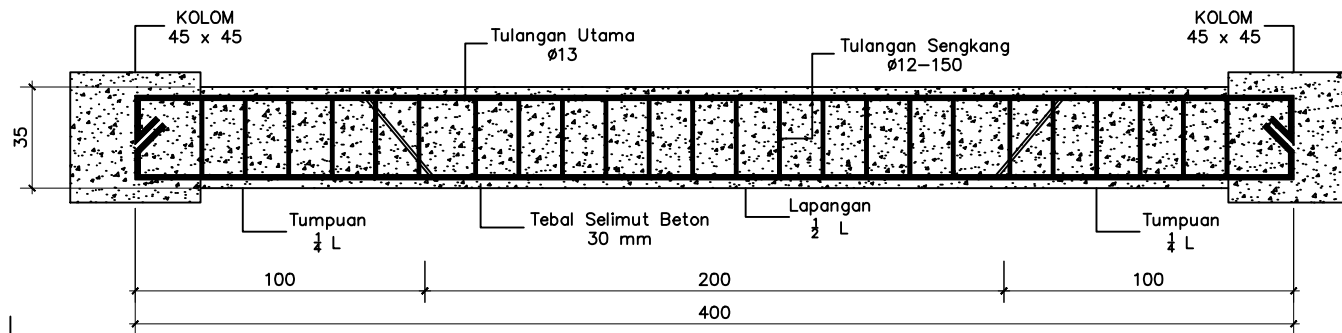


Potongan balok induk Arah X
SKALA 1 : 100

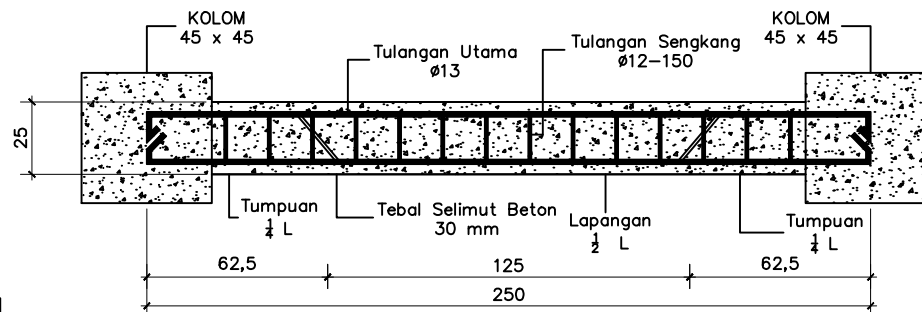


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NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : -		
NAMA GAMBAR : Potongan balok induk		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	11



Potongan balok induk Koridor Arah X
SKALA 1 : 100

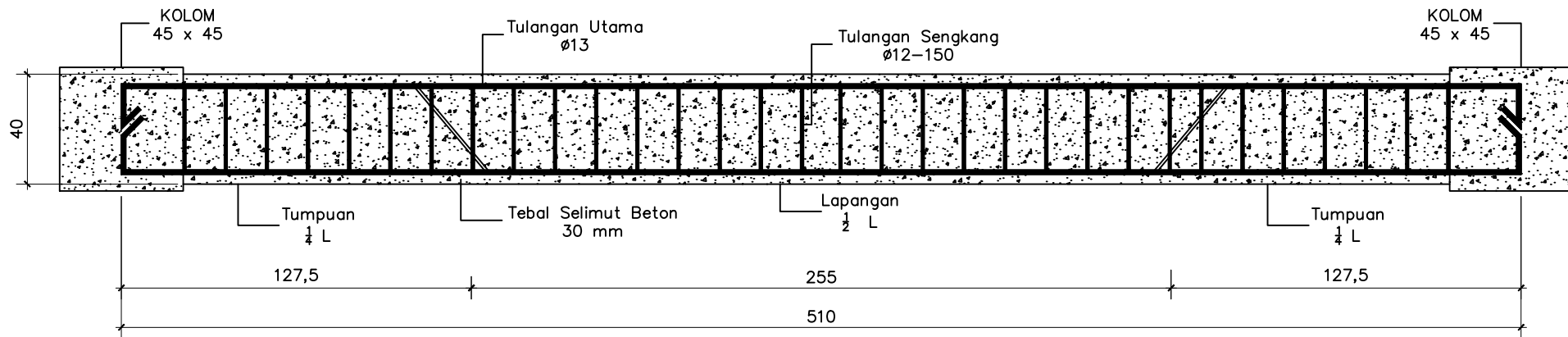


Potongan balok induk koridor Arah Y
SKALA 1 : 100

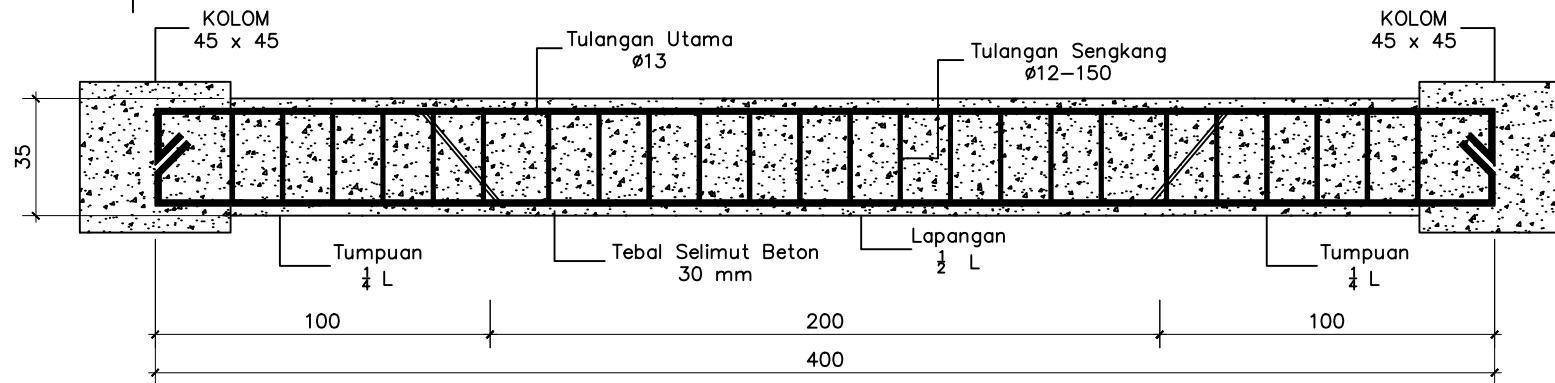


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(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR : Potongan balok Induk		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	12



Potongan Sloof Y
SKALA 1 : 100

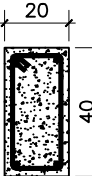
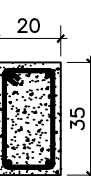
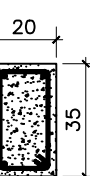


Potongan Sloof X
SKALA 1 : 100

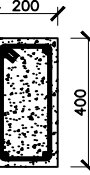
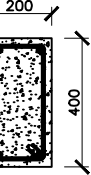


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(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR : Potongan Sloof		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	13

NAMA DETAIL	BALOK 1		BALOK 2	
	Tumpuan	Lapangan	Tumpuan	Lapangan
EL ±4.00 Lantai 2				
	20	20	20	20
	40	40	35	35
	2 # 13	2 # 13	2 # 13	2 # 13
Tulangan Tumpuan	2 # 13		2 # 13	
Tulangan Lapangan	2 # 13		2 # 13	
Jarak Sengkang	# 10 – 150		# 10 – 150	



NAMA DETAIL	SLOOF X		SLOOF Y	
	Tumpuan	Lapangan	Tumpuan	Lapangan
EL ±0.00				
	200	200	200	200
	400	400	400	400
	2 # 13	2 # 13	2 # 13	2 # 13
Tulangan Tumpuan	2 # 13		2 # 13	
Tulangan Lapangan	2 # 13		2 # 13	
Jarak Sengkang	# 10 – 150		# 10 – 150	

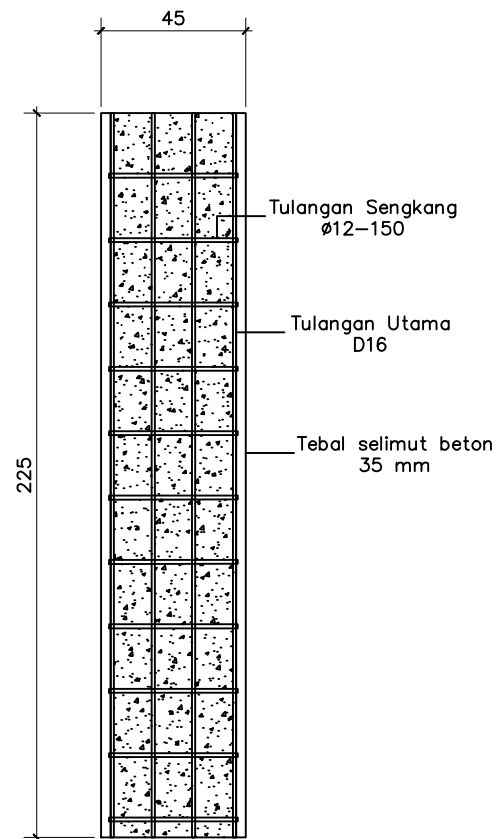


NAMA DETAIL	BALOK 3	
	Tumpuan	Lapangan
EL ±4.00 Lantai 2		
	15	15
	25	25
	2 # 13	2 # 13
Tulangan Tumpuan	2 # 13	
Tulangan Lapangan	2 # 13	
Jarak Sengkang	# 10 – 150	



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(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR :Detail balok dan Sloof		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	14



NAMA DETAIL	K1	K2
DETAIL KOLOM		
Tulangan Utama	8 D 16	8 D 16
Tulangan Sengkang	Ø12-150	Ø12-150

Detail Kolom
SKALA 1 : 20

Potongan Kolom
SKALA 1 : 100



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GUNUNG JATI CIREBON
(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1

ANGKATAN/KELAS : 2020/3A

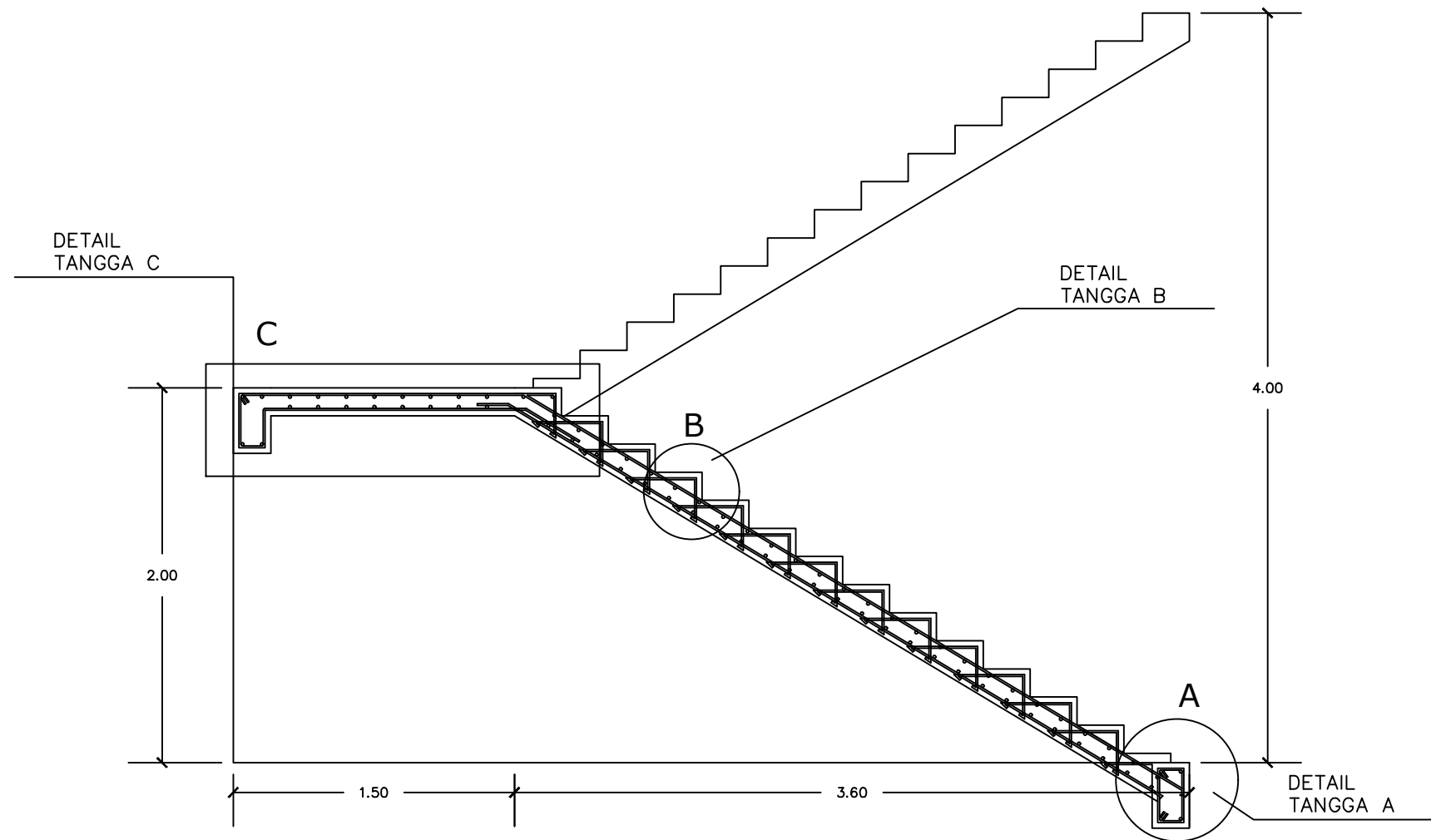
DOSEN : Aryati Indah.K.ST.,MT

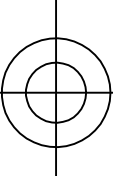
ASISTEN DOSEN : Ari al'fiansyah

FUNGSI BANGUNAN : —

NAMA GAMBAR : Potongan dan detail kolom

Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	15

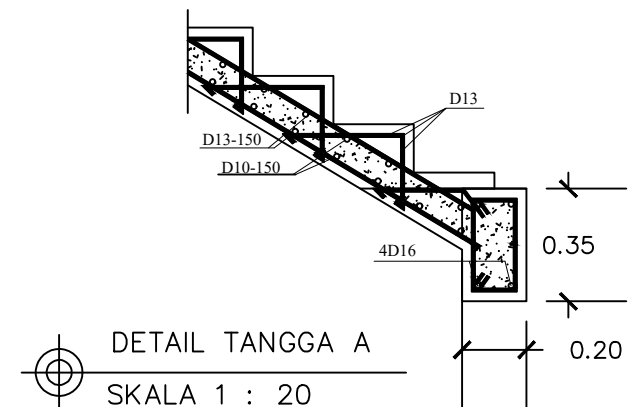
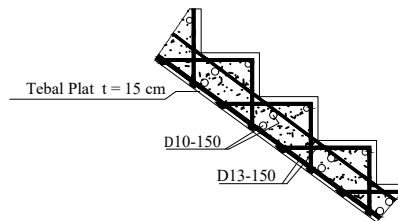
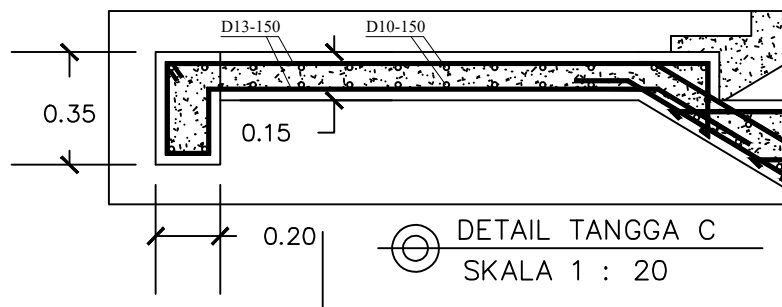



 Potongan Tulangan Tangga
 SKALA 1 : 100



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 (TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : –		
NAMA GAMBAR : Potongan Tangga		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	16



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GUNUNG JATI CIREBON
(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1

ANGKATAN/KELAS : 2020/3A

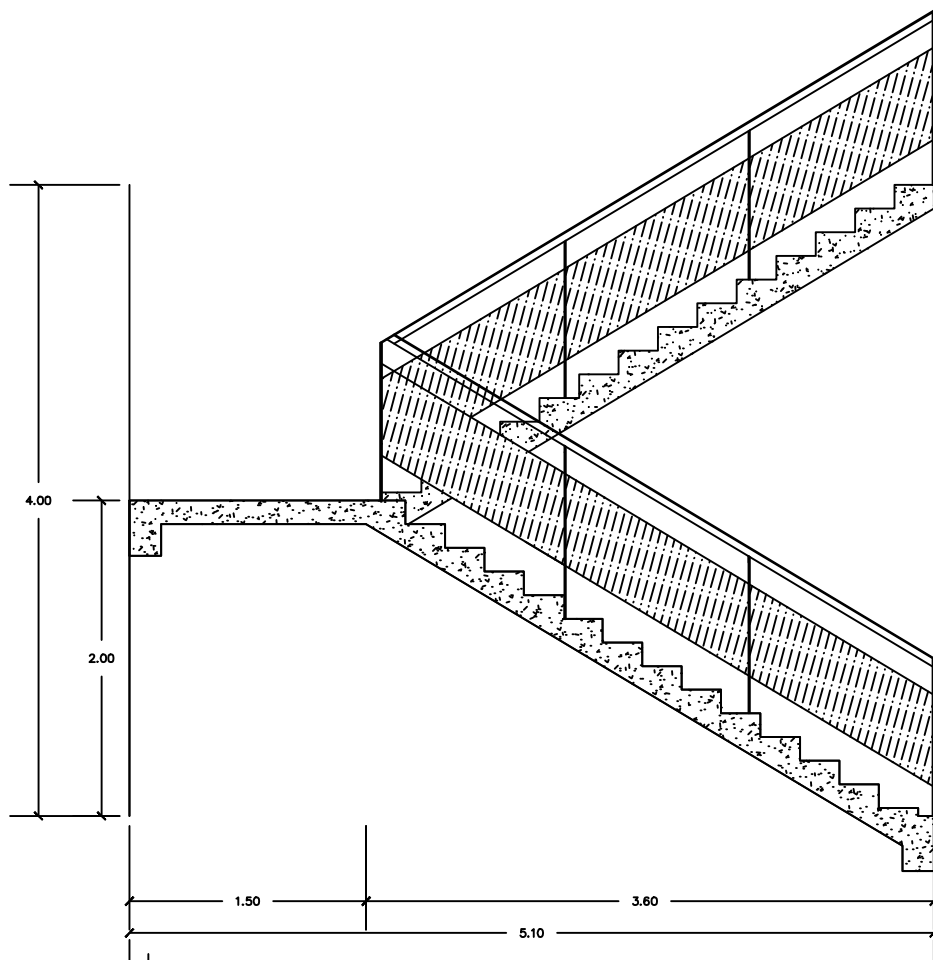
DOSEN : Aryati Indah.K.ST.,MT

ASISTEN DOSEN : Ari al'fiansyah

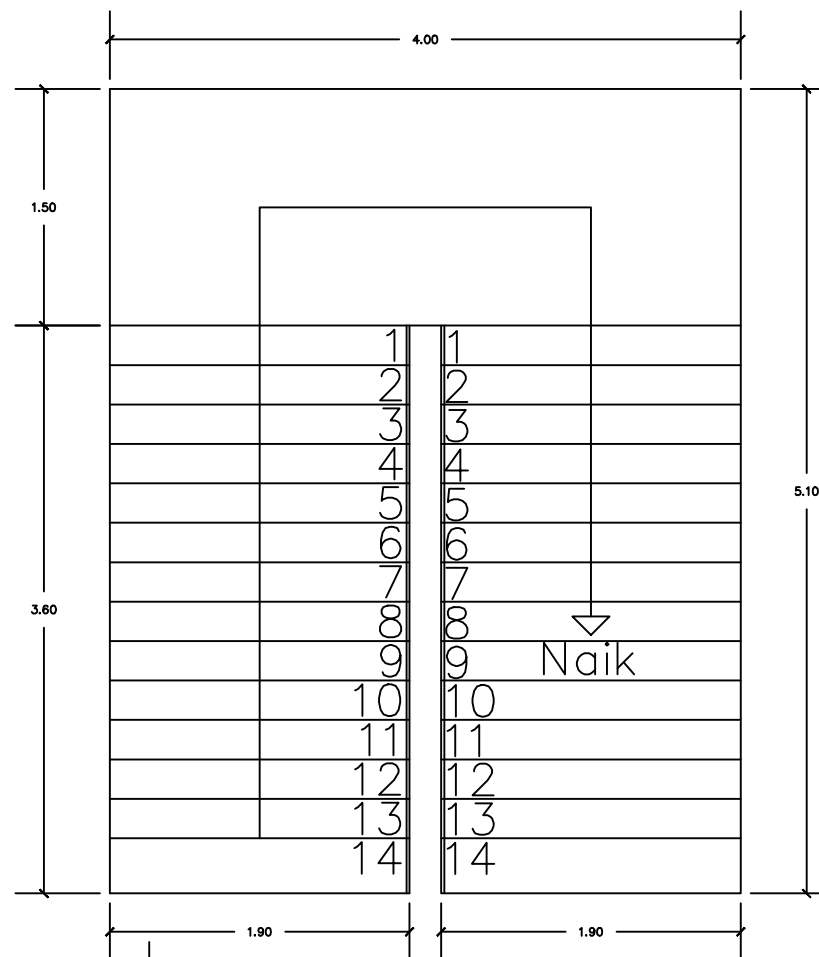
FUNGSI BANGUNAN : -

NAMA GAMBAR : Detail Tangga

Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	17



Model Tangga
SKALA 1 : 100

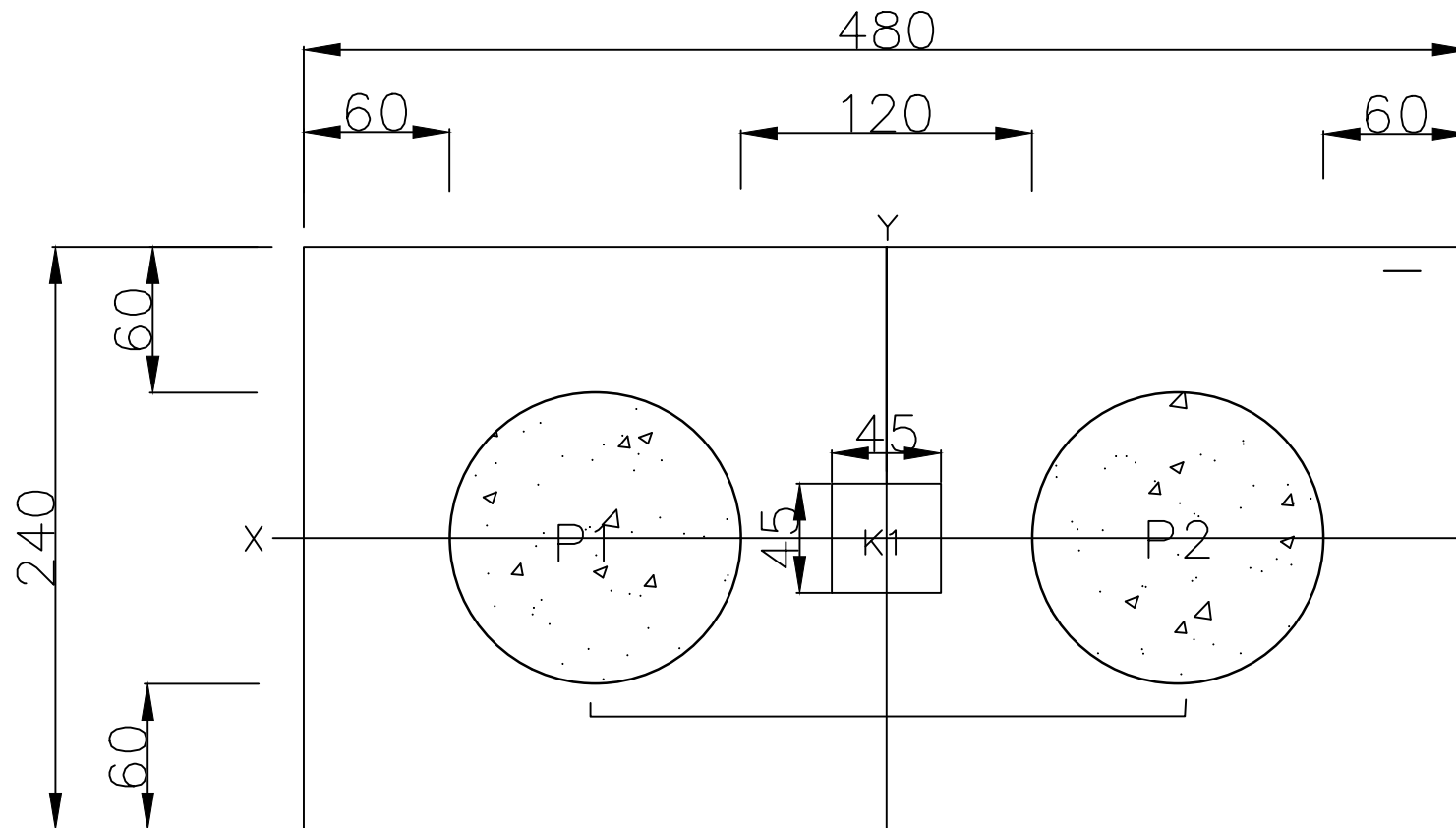


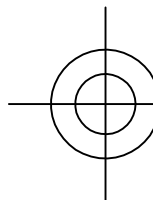
denah Tangga
SKALA 1 : 100



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(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1		
ANGKATAN/KELAS : 2020/3A		
DOSEN : Aryati Indah.K.ST.,MT		
ASISTEN DOSEN : Ari al'fiansyah		
FUNGSI BANGUNAN : -		
NAMA GAMBAR : Denah dan Model Tangga		
Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	18




 Denah Tiang Pancang Grup
 SKALA 1 : 100



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 (TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1

ANGKATAN/KELAS : 2020/3A

DOSEN : Aryati Indah.K.ST.,MT

ASISTEN DOSEN : Ari al'fiansyah

FUNGSI BANGUNAN : –

NAMA GAMBAR : Denah Tiang Pancang

Tanggal

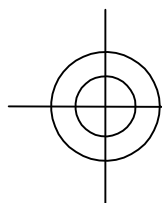
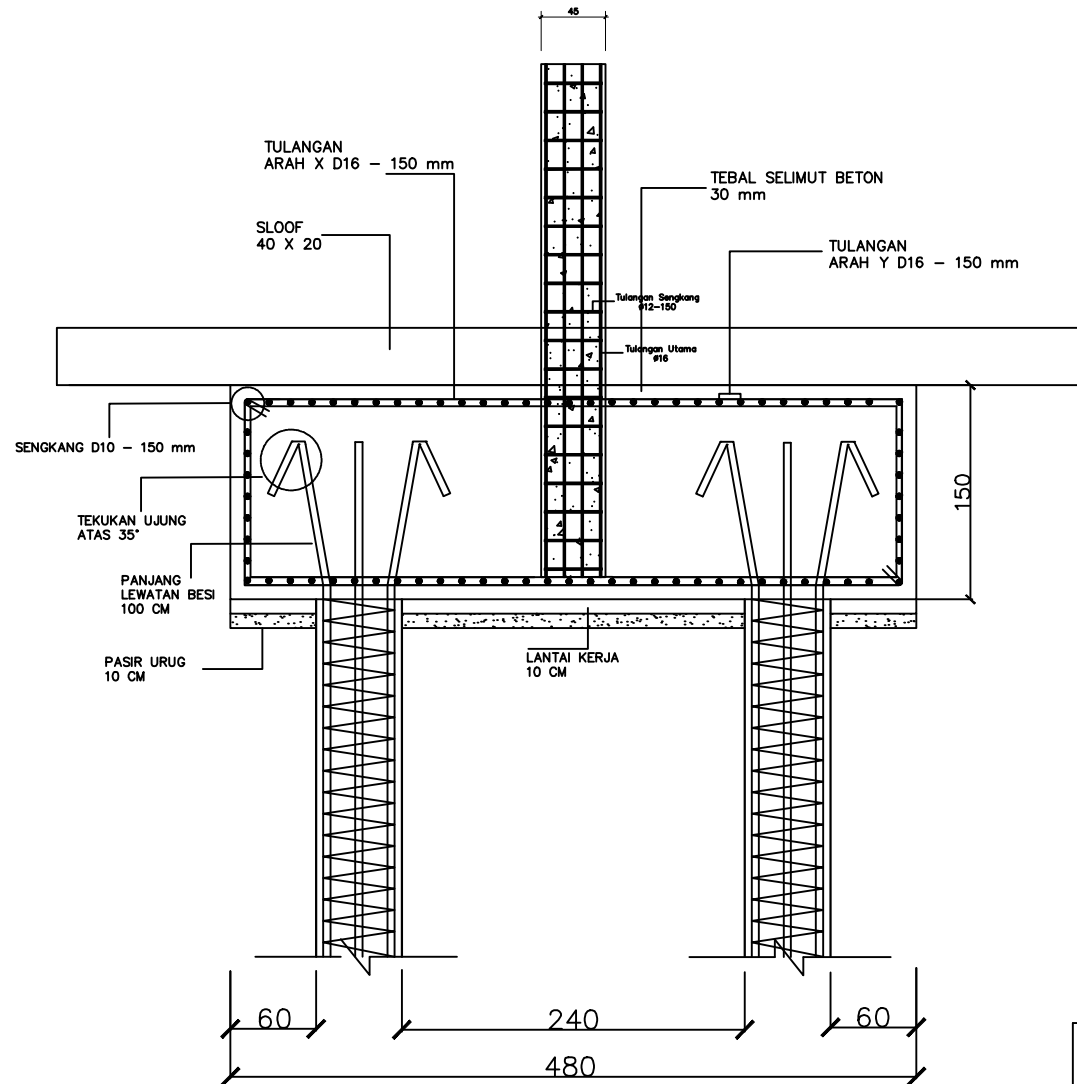
Skala

No.Draw

31 Juli 2023

1 : 100

19



Potongan Pile Cap & Pancang

SKALA 1 : 100



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(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1

ANGKATAN/KELAS : 2020/3A

DOSEN : Aryati Indah.K.ST.,MT

ASISTEN DOSEN : Ari al'fiansyah

FUNGSI BANGUNAN : -

NAMA GAMBAR : Potongan Pile Cap & Pancang

Tanggal

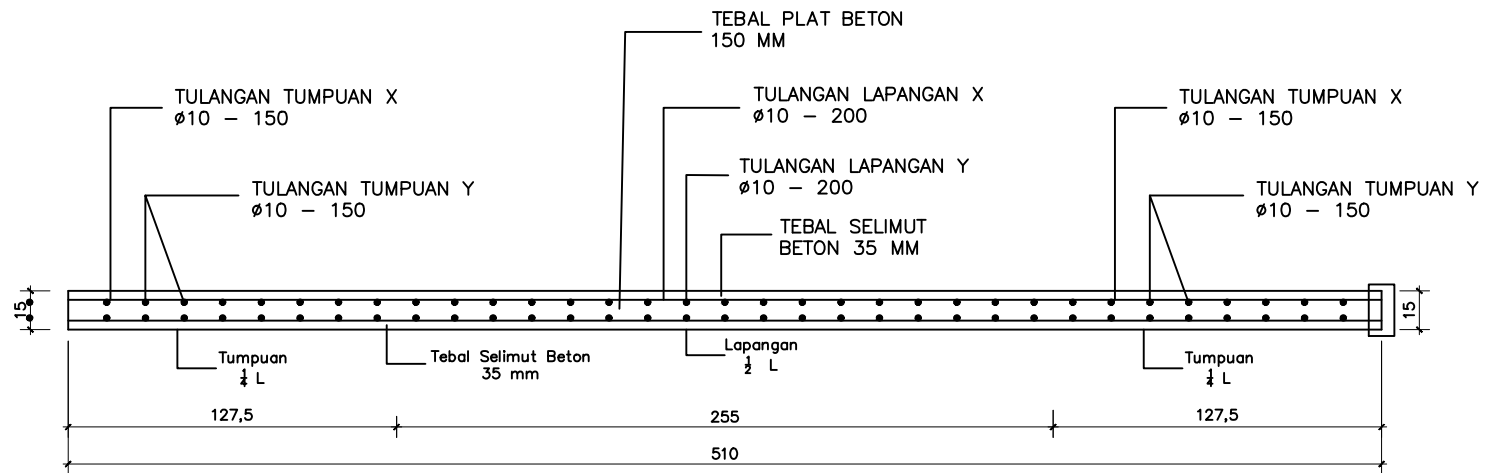
Skala

No.Draw

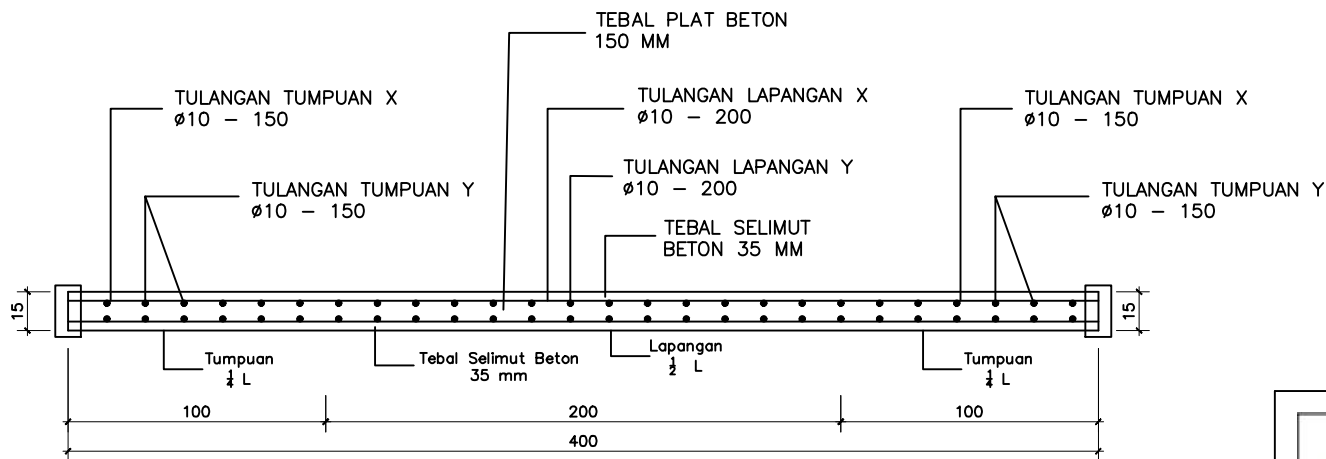
31 Juli 2023

1 : 100

20



Potongan Floor Slab X LT 2-4
SKALA 1 : 100



Potongan Sloof arah Y LT 2-4
SKALA 1 : 100



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(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1

ANGKATAN/KELAS : 2020/3A

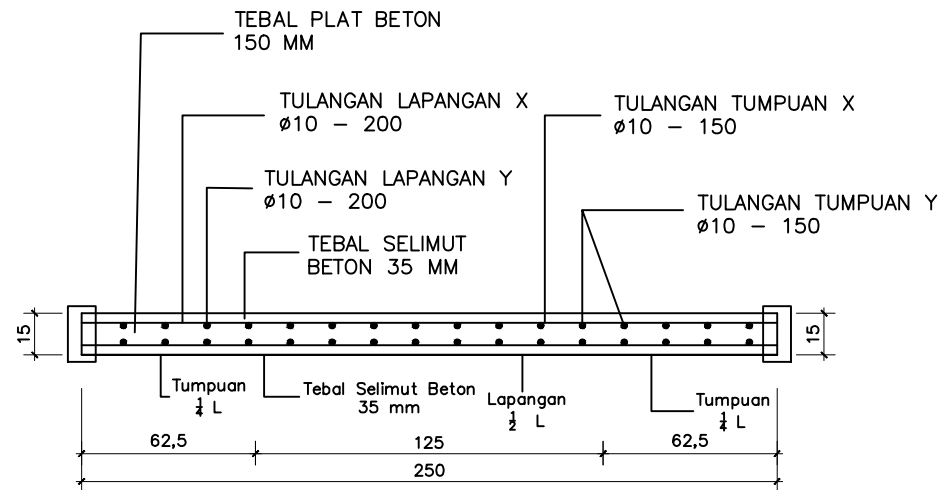
DOSEN : Aryati Indah.K.ST.,MT

ASISTEN DOSEN : Ari al'fiansyah

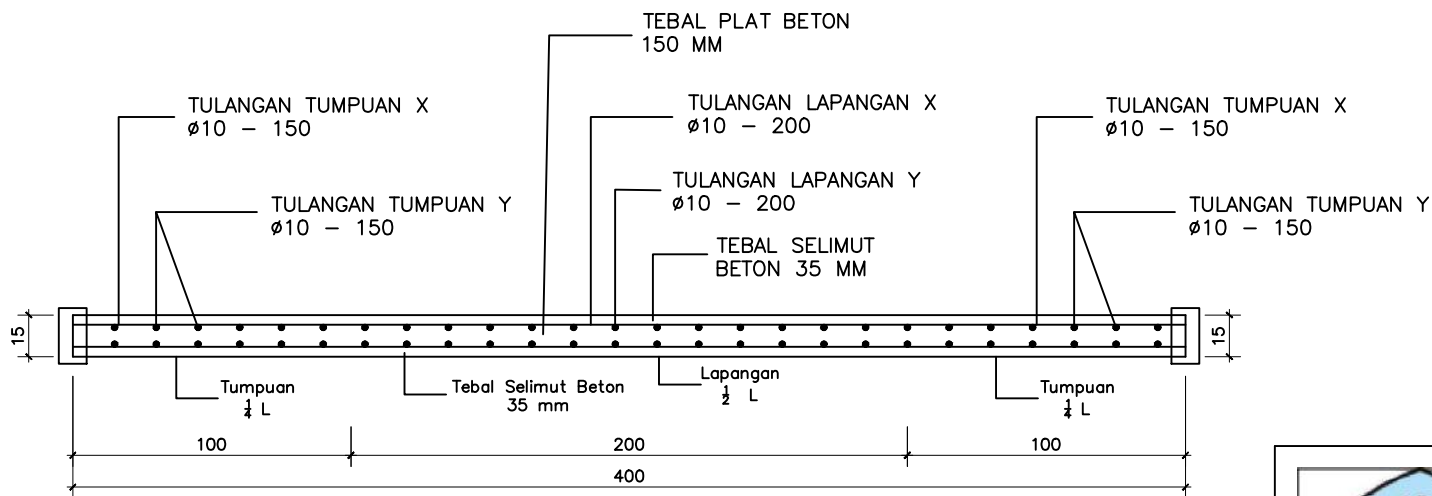
FUNGSI BANGUNAN : -

NAMA GAMBAR : Potongan Slab

Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	21



Potongan Sloof arah Y LT 2-4 (Koridor)
SKALA 1 : 100



Potongan Sloof arah X LT 2-4 (Koridor)
SKALA 1 : 100



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(TEKNIK SIPIL)

NAMA KELOMPOK : Kelompok 1

ANGKATAN/KELAS : 2020/3A

DOSEN : Aryati Indah.K.ST.,MT

ASISTEN DOSEN : Ari al'fiansyah

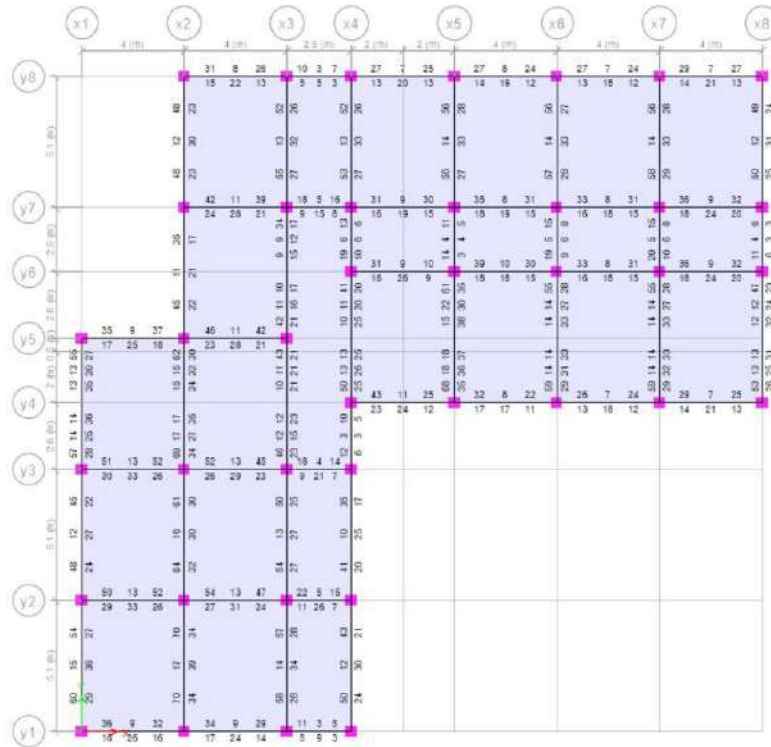
FUNGSI BANGUNAN : -

NAMA GAMBAR : Potongan Slab Koridor

Tanggal	Skala	No.Draw
31 Juli 2023	1 : 100	22

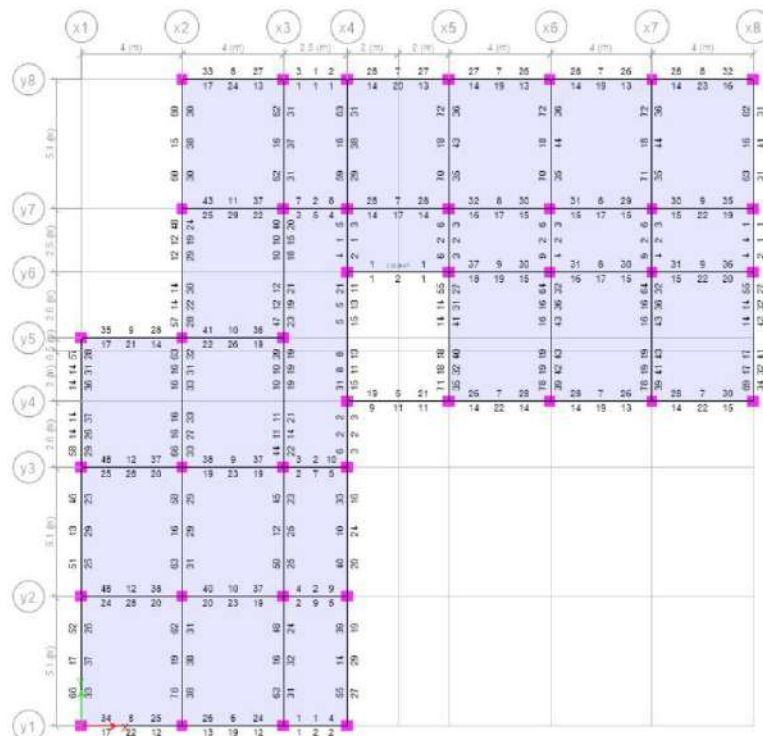
LAMPIRAN

Plan View - Story1 - Z = 2 (m) Longitudinal Reinforcing (ACI 318-14)



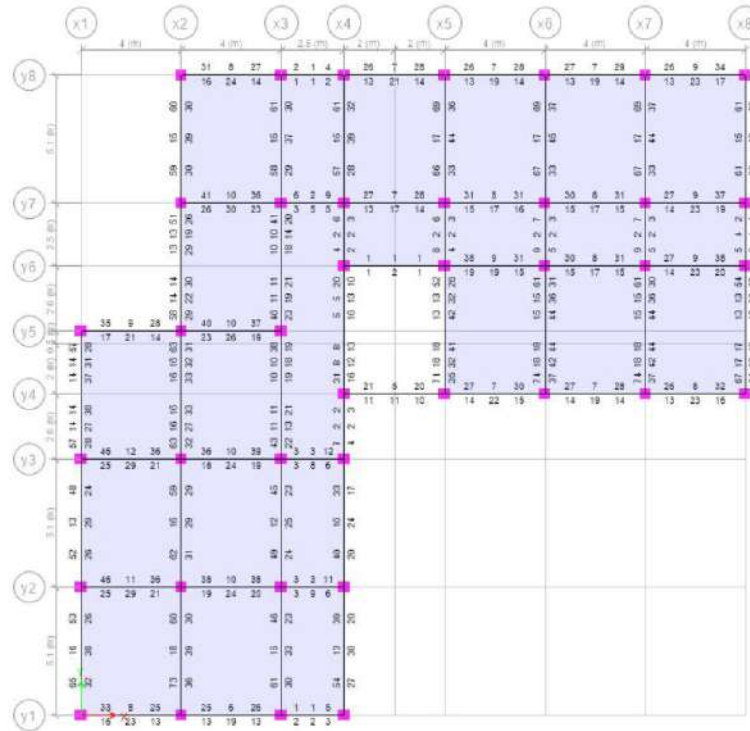
ETABS Lantai 1

Plan View - Story2 - Z = 6 (m) Longitudinal Reinforcing (ACI 318-14)



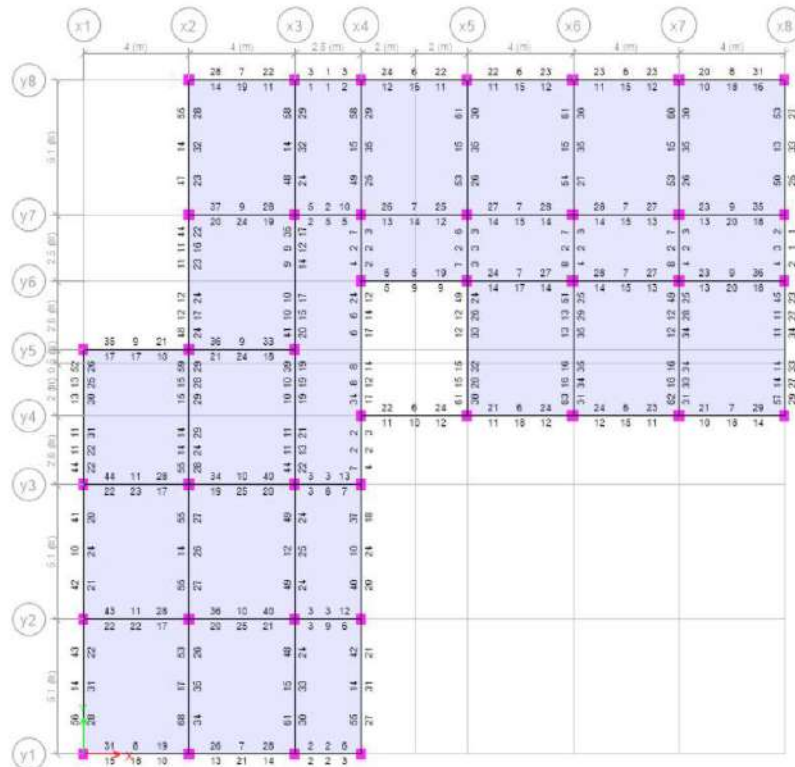
ETABS Lantai 2

Plan View - Story3 - Z = 10 (m) Longitudinal Reinforcing (ACI 318-14)

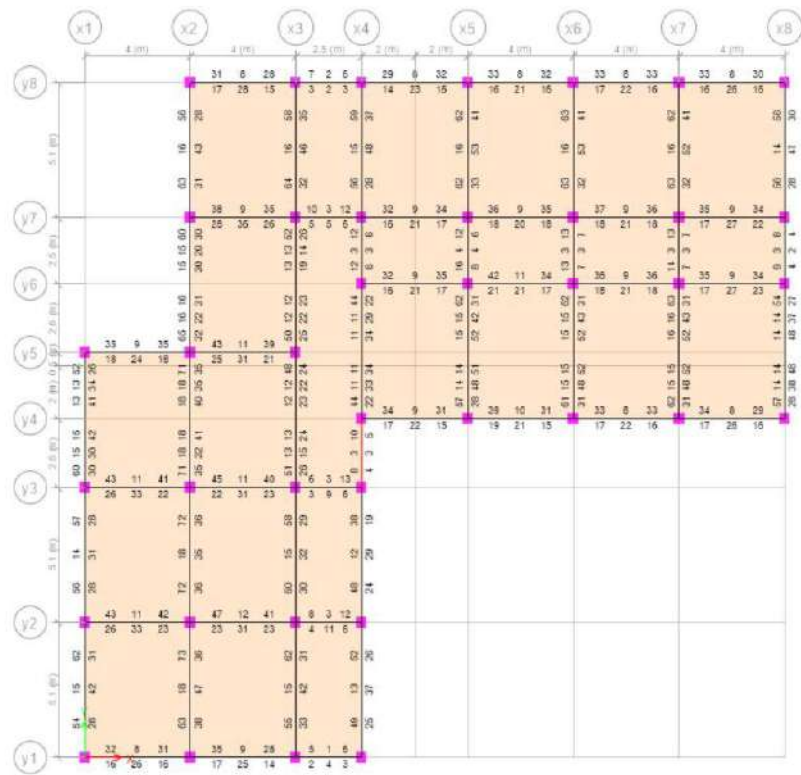


ETABS Lantai 3

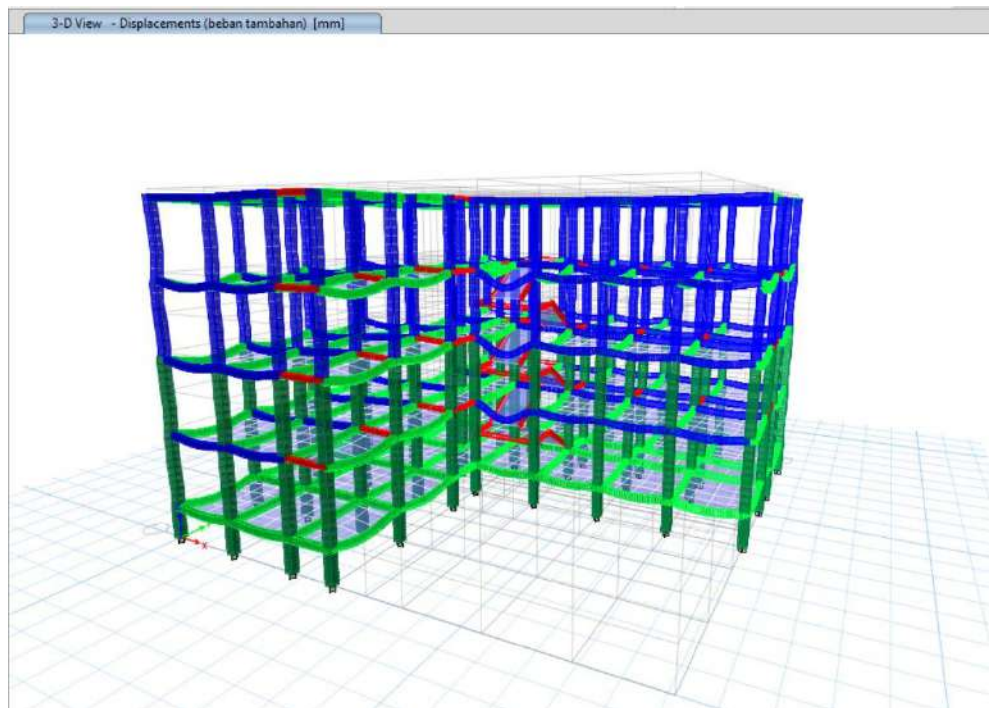
Plan View - Story4 - Z = 14 (m) Longitudinal Reinforcing (ACI 318-14)



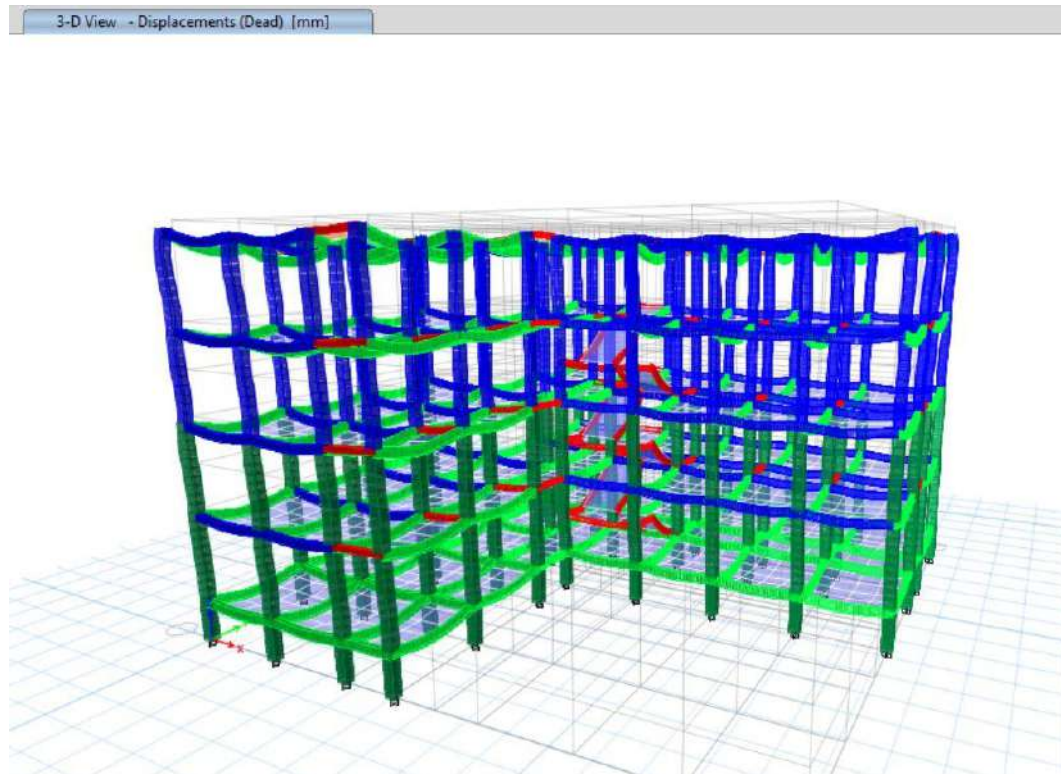
ETABS Lantai 4



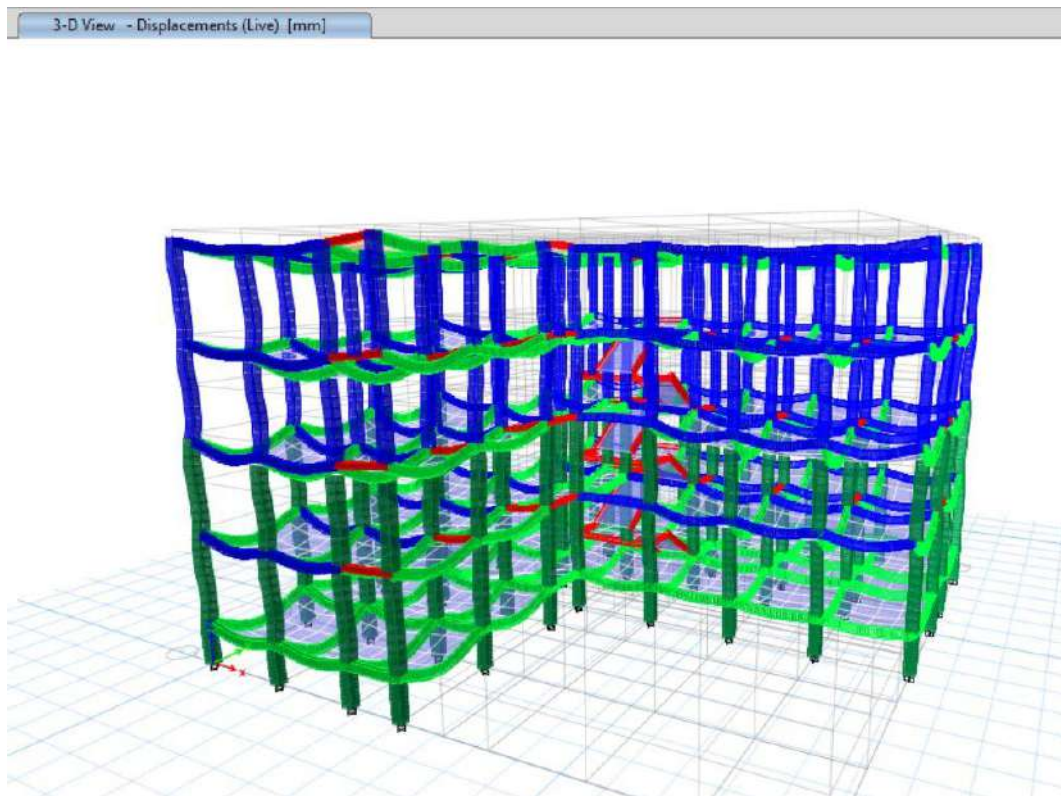
ETABS Atap



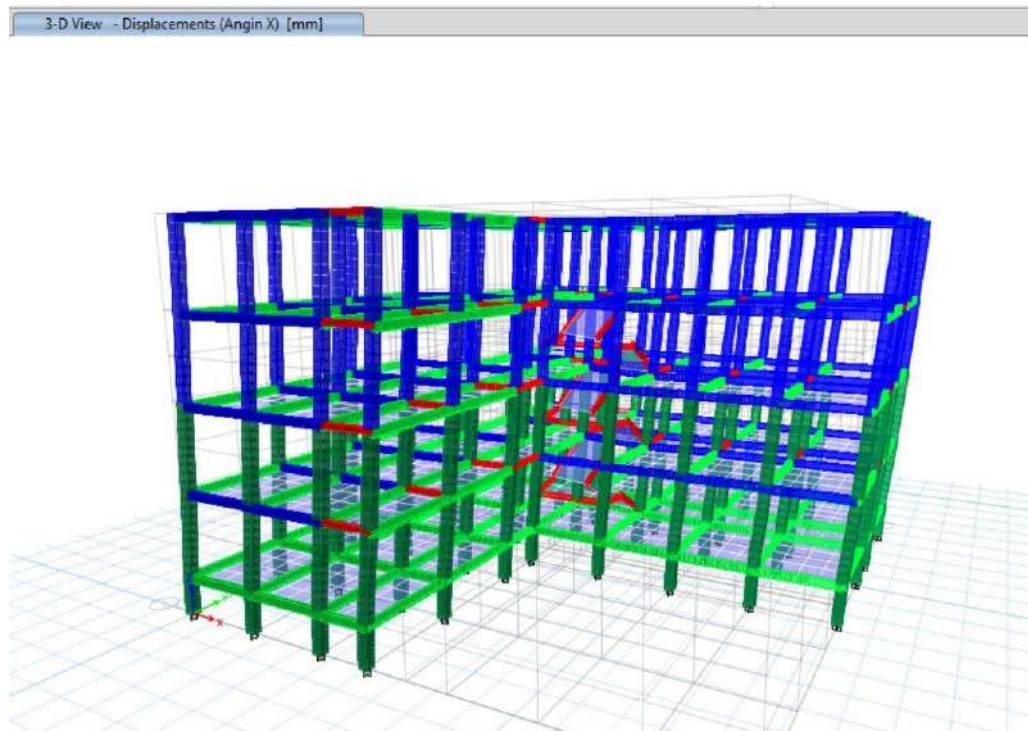
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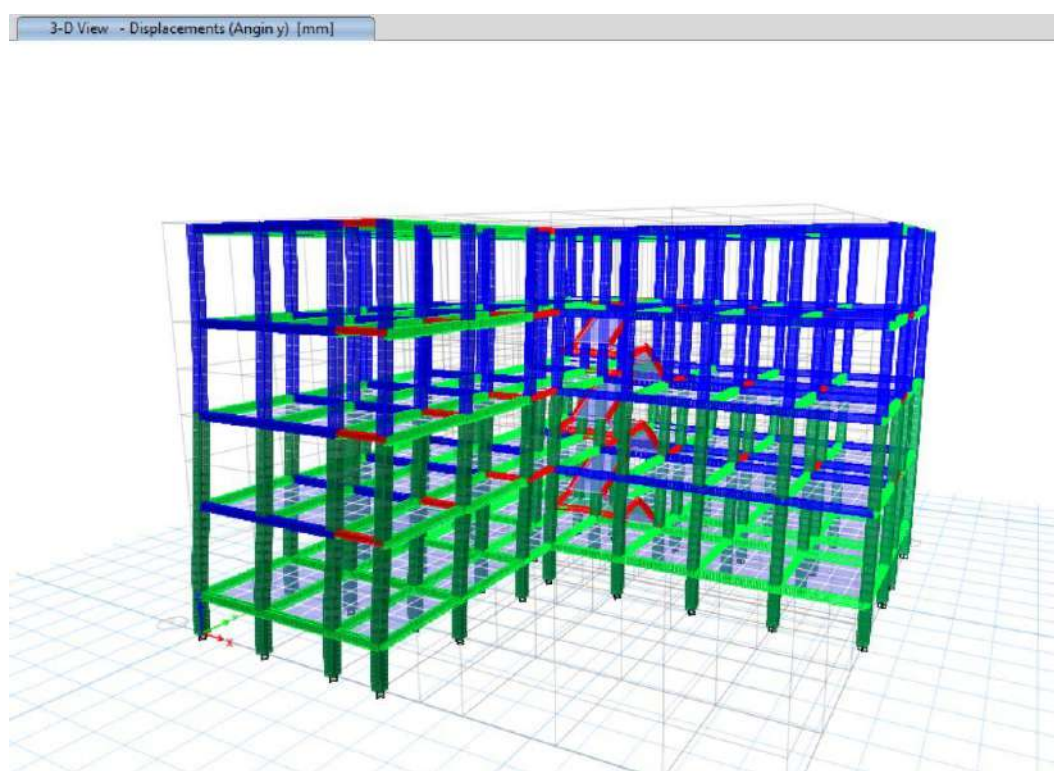
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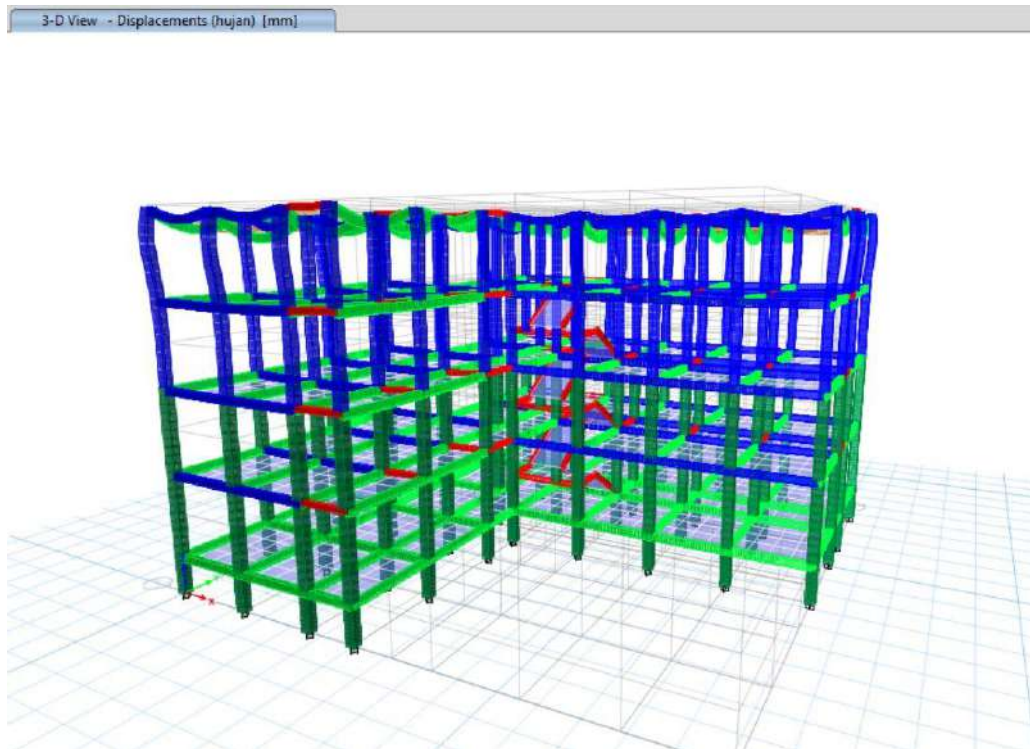
ETABS Beban Hidup



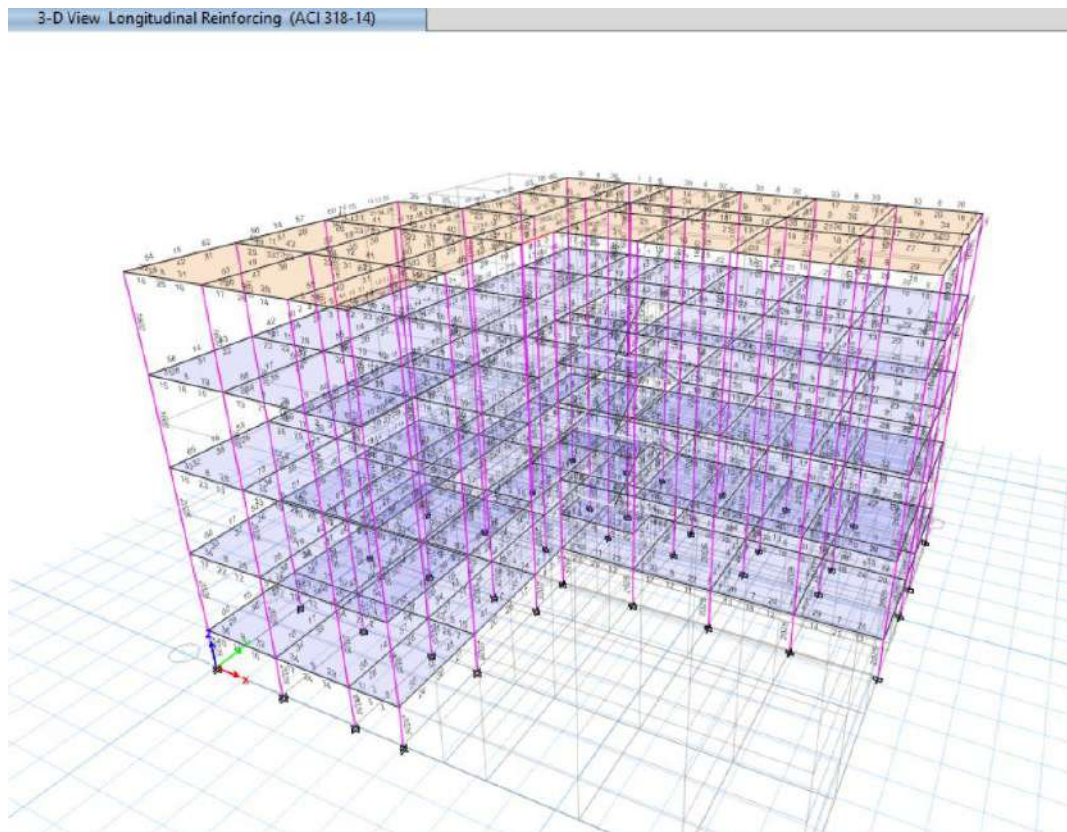
ETABS Beban angin X



ETABS Beban Angin Y



ETABS Beban Hujan



Etabs Tampak 3 dimensi



Spektrum Gempa cirebon

Beam Element Details (Summary)

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
Story2	B44	515	B1	1.2 DL + 1 EQY + 1 LL	225	2500	1	Sway Special



Section Properties

b (mm)	h (mm)	b _x (mm)	d _x (mm)	d _y (mm)	d _{xy} (mm)
200	400	200	0	30	30

Material Properties

E _c (MPa)	f _c (MPa)	Lt.Wt.Factor (Unitless)	f _y (MPa)	f _w (MPa)
25742.98	30	1	545	405

Design Code Parameters

ϕ_c	$\phi_{c,LR}$	$\phi_{c,LR,LR}$	$\phi_{t,LR}$	$\phi_{t,LR}$	$\phi_{w,LR}$
0.9	0.65	0.75	0.75	0.6	0.85

Design Moment and Flexural Reinforcement for Moment, M_{u3}

	Design Moment kN-m	Design P _u kN	-Moment Rebar mm ²	+Moment Rebar mm ²	Minimum Rebar mm ²	Required Rebar mm ²
Top (+2 Axis)	-10.4609	-0.0001	58	0	78	78
Bottom (-2 Axis)	5.2305	-0.0001	0	29	39	39

Shear Force and Reinforcement for Shear, V_{u2}

Shear V _{u2} kN	Shear ϕV_c kN	Shear ϕV_s kN	Shear V _u kN	Rebar A _v / S mm ² /m
25.0943	50.4828	0	8.2851	0

Torsion Force and Torsion Reinforcement for Torsion, T_u

T _u kN-m	ϕT_c kN-m	ϕT_s kN-m	Area A _c cm ²	Perimeter, P _u mm	Rebar A _t / s mm ² /m	Rebar A _t mm ²
0.0071	1.8192	7.2768	293.8	844.4	0	0

Detail Etabs B1

Column Element Details (Summary)								
Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
Story2	C47	452	K1	1.2 DL + 1.6 WY + 1 LL + 0.5 R	0	4000	0.5	Sway Special

Section Properties				
b (mm)	h (mm)	dc (mm)	Cover (Torsion) (mm)	
450	450	58	27.3	

Material Properties				
E_c (MPa)	f'_c (MPa)	Lt.Wt. Factor (Unitless)	f_y (MPa)	f_{wy} (MPa)
25742.96	30	1	545	405

Design Code Parameters						
ϕ_c	$\phi_{c,wy}$	$\phi_{c,trans}$	ϕ_{wy}	ϕ_{yx}	$\phi_{w,wy}$	Ω_u
0.9	0.65	0.75	0.75	0.6	0.85	2

Axial Force and Biaxial Moment Design For P_{u2} , M_{u2}						
Design P_u kN	Design M_{u2} kN-m	Design M_{u3} kN-m	Minimum M2 kN-m	Minimum M3 kN-m	Rebar Area mm ²	Rebar % %
939.1085	26.99	-1.5026	26.99	26.99	2025	1

Axial Force and Biaxial Moment Factors					
	C_m Factor Unitless	δ_{ns} Factor Unitless	δ_x Factor Unitless	K Factor Unitless	Effective Length mm
Major Bend(M3)	0.265704	1	1	1	3600
Minor Bend(M2)	0.416754	1	1	1	3600

Shear Design for V_{u2} , V_{u3}					
	Shear V_u kN	Shear ϕV_u kN	Shear ϕV_u kN	Shear ϕV_u kN	Rebar A_v /s mm ² /m
Major, V_{u2}	0.7662	160.8124	0	0	0
Minor, V_{u3}	0.1734	160.8124	0	6.3053	0

Detail Etabs Kolom 1

Beam Element Details (Summary)								
Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
tangga 3	D1	201	B3	1.4 DL	0	3689.2	1	Sway Special

Section Properties						
b (mm)	h (mm)	b _f (mm)	d _s (mm)	d _{cs} (mm)	d _{cs} (mm)	
150	250	150	0	30	30	

Material Properties				
E_c (MPa)	f'_c (MPa)	Lt.Wt. Factor (Unitless)	f_y (MPa)	f_{wy} (MPa)
25742.96	30	1	545	405

Design Code Parameters						
ϕ_c	$\phi_{c,wy}$	$\phi_{c,trans}$	ϕ_{wy}	ϕ_{yx}	$\phi_{w,wy}$	
0.9	0.65	0.75	0.75	0.6	0.85	

Design Moment and Flexural Reinforcement for Moment, M_{u3}						
	Design Moment kN-m	Design P_u kN	-Moment Rebar mm ²	+Moment Rebar mm ²	Minimum Rebar mm ²	Required Rebar mm ²
Top (+2 Axis)	-2.4007	7.368	14	0	19	19
Bottom (-2 Axis)	1.2004	7.368	0	3	4	4

Shear Force and Reinforcement for Shear, V_{u2}					
Shear V_{u2} kN	Shear ϕV_u kN	Shear ϕV_u kN	Shear V_u kN	Rebar A_v /s mm ² /m	
3.6914	22.5126	0	1.0369	0	

Torsion Force and Torsion Reinforcement for Torsion, T_u						
T_u kN-m	ϕT_u kN-m	ϕT_u kN-m	Area A_c cm ²	Perimeter, p_h mm	Rebar A_t /s mm ² /m	Rebar A_t mm ²
0.0015	0.6311	2.5246	83.7	444.4	0	0

Detail Etabs Tangga

Column Element Details (Summary)

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
Story1	C25	48	K1	1.2 DL + 1.6 WY + 1 LL + 0.5 R	0	2000	0.5	Sway Special

Section Properties

b (mm)	h (mm)	dc (mm)	Cover (Torsion) (mm)
450	450	58	27.3

Material Properties

E_c (MPa)	f'_c (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
25742.98	30	1	545	405

Design Code Parameters

ϕ_T	ϕ_{CTes}	ϕ_{CSoln}	ϕ_{Ves}	ϕ_{Ves}	ϕ_{Vesint}	Ω_0
0.9	0.85	0.75	0.75	0.6	0.85	2

Axial Force and Biaxial Moment Design For P_u , M_{u2} , M_{u3}

Design P_u kN	Design M_{u2} kN-m	Design M_{u3} kN-m	Minimum M2 kN-m	Minimum M3 kN-m	Rebar Area mm ²	Rebar % %
1211.5807	1.1337	-34.8208	34.8208	34.8208	2025	1

Axial Force and Biaxial Moment Factors

	C_m Factor Unitless	δ_{ex} Factor Unitless	δ_{ey} Factor Unitless	K Factor Unitless	Effective Length mm
Major Bend(M3)	0.818548	1	1	1	1800
Minor Bend(M2)	0.437374	1	1	1	1800

Shear Design for V_{u2} , V_{u3}

	Shear V_u kN	Shear ϕV_u kN	Shear ϕV_u kN	Shear ϕV_u kN	Rebar A_v /s mm ² /m
Major, V_{u2}	0.2225	172.5549	0	0	0
Minor, V_{u3}	2.4533	172.5549	0	19.9781	0

Detail Etabs Pondasi

Beam Element Details (Summary)

Level	Element	Unique Name	Section ID	Combo ID	Station Loc	Length (mm)	LLRF	Type
tangga 3	B13	197	B3	1.4 DL	0	2000	1	Sway Special

Section Properties

b (mm)	h (mm)	b_f (mm)	d_s (mm)	d_{ab} (mm)	d_{ab} (mm)
150	250	150	0	30	30

Material Properties

E_c (MPa)	f'_c (MPa)	Lt.Wt Factor (Unitless)	f_y (MPa)	f_{ys} (MPa)
25742.98	30	1	545	405

Design Code Parameters

ϕ_T	ϕ_{CTes}	ϕ_{CSoln}	ϕ_{Ves}	ϕ_{Ves}	ϕ_{Vesint}
0.9	0.85	0.75	0.75	0.6	0.85

Design Moment and Flexural Reinforcement for Moment, M_{u3}

	Design Moment kN-m	Design P_u kN	-Moment Rebar mm ²	+Moment Rebar mm ²	Minimum Rebar mm ²	Required Rebar mm ²
Top (+2 Axis)	-1.9731	1.6289	17	0	22	22
Bottom (-2 Axis)	0.9865	1.6289	0	7	10	10

Shear Force and Reinforcement for Shear, V_{u2}

Shear V_{u2} kN	Shear ϕV_u kN	Shear ϕV_u kN	Shear V_u kN	Rebar A_v /S mm ² /m
3.2195	22.5126	0	1.8523	0

Torsion Force and Torsion Reinforcement for Torsion, T_u

T_u kN-m	ϕT_u kN-m	ϕT_u kN-m	Area A_o cm ²	Perimeter, P_o mm	Rebar A_t /s mm ² /m	Rebar A_t mm ²
0.1446	0.8087	2.4288	83.7	444.4	0	0

Detail Etabs Bordes tangga